TECHNICAL REPORT

# VLMProbe:

## An Open-source Library for VLM Probing and Interpretation

Exploring Model Internals Beyond the Answer

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**ABSTRACT**

Understanding how vision-language models use visual information requires connecting internal representations to model behavior. We present vlmprobe, an open-source library that integrates lens readouts, attention attribution, causal interventions, and sparse autoencoder (SAE) analysis within a common model interface. Shared adapters, calibration and checkpoint utilities, evaluation pipelines, and visualizations support systematic analysis across architectures. Using this framework, we compare methods across vision-language, language, and vision models through matched case studies and controlled evaluations of pretrained sparse dictionaries. The comparisons show that representation readouts and spatial explanations depend on the analysis objective and observation site, and that local attribution estimates can disagree with measured intervention effects. SAE evaluations further expose distinct trade-offs in reconstruction fidelity, concept readout, and spatial localization. Together, these results support combining multiple forms of evidence to evaluate an interpretation. vlmprobe provides a reusable basis for selecting methods, comparing their conclusions, and testing hypotheses about model behavior.

## 1 Introduction

Lens readouts, attention attribution, causal interventions, and sparse features offer complementary views of a vision-language model (Belrose et al. 2023; Chefer et al. 2021; Wang et al. 2023; Gao, Tour, et al. 2024). Their outputs answer different questions, while separate implementations make consistent comparison across architectures difficult.

vlmprobe integrates these four families in a shared workflow for pretrained vision-language, language, and vision models. Adapters preserve native activation sites and token layouts, while common interfaces support execution and result storage. Calibration, checkpoint loading, evaluation, and visualization utilities let users combine methods without rebuilding model-specific analysis code.

Using the library, we conduct matched VLM case studies and complementary language and vision comparisons. Inputs and semantic targets are fixed within compatible comparisons, with each method retaining its analysis unit. Controlled pretrained-SAE evaluations examine reconstruction, concept readout, and spatial localization.

The comparisons show that readouts and spatial maps depend on the analysis objective and observation site. Local attribution estimates can disagree with measured intervention effects, while SAE reconstruction and feature utility require separate evaluation. These findings support selecting methods for a specific question, combining their evidence, and testing explanatory hypotheses against model behavior.

# 2 Library and evaluation protocol

## 2.1 Model adapters and method configuration

Figure 1 summarizes the library workflow. An input contains the native model tensors and their text/image coordinates. Configuration specifies the method, activation sites, positions, and target score; paired inputs and calibrated artifacts are supplied where required. Outputs retain these choices alongside decoded tokens, attribution maps, or intervention effects. This shared contract allows methods to reuse an input preparation step and a visualization without assuming identical internal architectures.

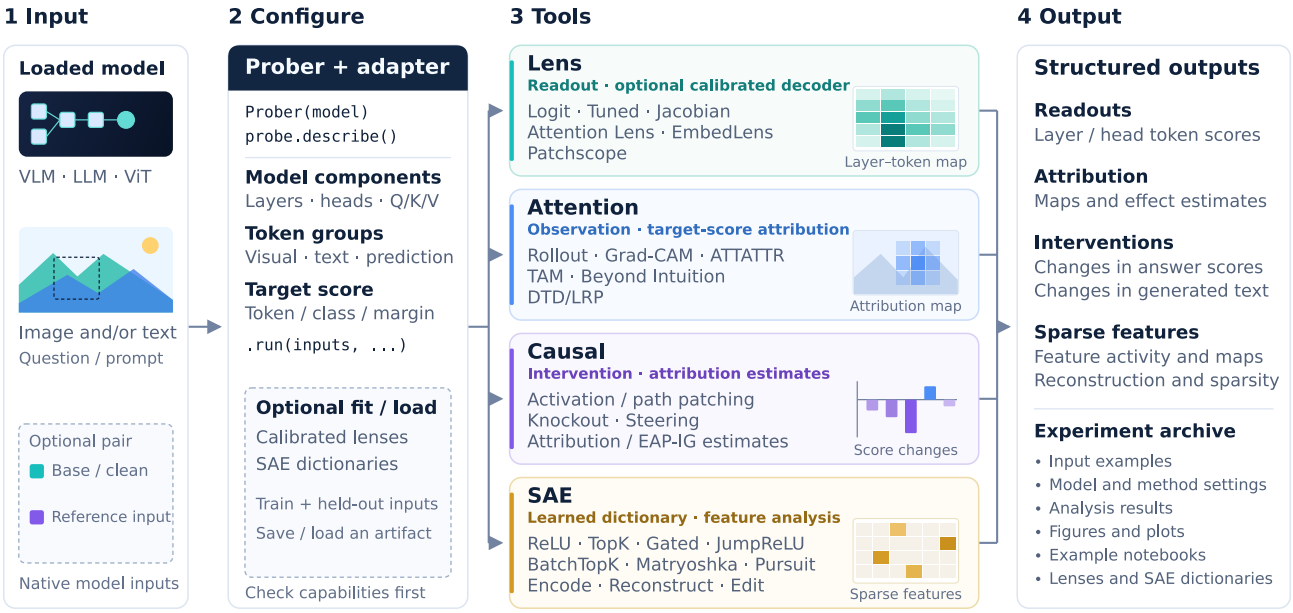



**Figure 1.** The vlmprobe workflow. Native inputs and explicit analysis settings feed lens, attention, causal and sparse-feature methods; their outputs support token readouts, spatial explanations, sparse feature analysis and measured interventions. Miniature plots illustrate output types rather than experimental results.

The following example reads a representation, explains an answer with attention gradients, and compares exact and first-order residual replacement:

```
from vlmprobe import Prober
from vlmprobe.metrics import SequenceLogProb

probe = Prober(model, processor)
metric = SequenceLogProb(answer_mask)

readout = probe.lens.logit(
    layers=layers, tokens=prediction_positions
).run(inputs)
attention = probe.attention.grad_cam(
    layers=attention_layers, queries=prediction_mask, keys="visual"
).run(inputs, metric=metric)
exact = probe.causal.patch(layers=layers, tokens="visual").run(
    inputs, source=reference, metric=metric
)
estimate = probe.causal.attribute(layers=layers, tokens="visual").run(
    inputs, source=reference, metric=metric
)
```

Here `inputs` includes the fixed answer for teacher-forced scoring; `answer_mask` selects its token pieces, and `reference` supplies aligned source activations. The method implementations share the adapter interface,

while `probe.describe()` exposes capabilities relevant to choosing sites and tools. The executed notebooks supply the native preprocessing and position-selection steps omitted from this compact example.

## 2.2 Models, inputs, and comparison groups

Table 1 lists the five pretrained VLMs used for the shared case studies. They vary in decoder architecture, visual-token compression, and model size. Table 2 defines the eight fixed inputs, chosen before inference. GQA supplies a common attribute question and target boxes (Hudson and Manning 2019); MMVP contrasts visually similar images (Tong et al. 2024); VSR tests spatial propositions (Liu et al. 2023); and VisOnlyQA supplies diagrams and charts (Kamoi et al. 2025).

**Table 1.** Models in the shared-case comparison. Sizes follow public checkpoint names. Native adapters retain each model's decoder sites and visual-token grid.

| Model | Image grid | Architectural feature |
|---|---|---|
| SmolVLM-500M-Instruct | $8 \times 8$ | Compressed image tokens with a Llama decoder |
| InternVL3-1B-hf | $16 \times 16$ | InternVL vision path with a Qwen2 decoder |
| Qwen3-VL-2B-Instruct | $14 \times 14$ | Normalized Q/K and DeepStack visual additions |
| Pixtral-12B | $16 \times 16$ | Pixtral patch encoder with a Mistral decoder |
| LLaVA-1.5-7B | $24 \times 24$ | CLIP visual features projected into a Llama decoder |

**Table 2.** Predetermined shared inputs. Each source contributes two cases; image captions from a separate COCO split calibrate the VLM Tuned Lenses.

| Source | Released identifiers | Analysis question |
|---|---|---|
| GQA | 07302654 / 19225230 | Is the shirt black or yellow? |
| MMVP | Images 1 / 2 | Are the butterfly's wings open or closed? |
| VSR zero-shot | Test rows 0 / 1 | Car behind suitcase; cat under backpack |
| VisOnlyQA real | Row 0 of two splits | Connected triangle; chart-value extraction |

For these shared cases, the same photograph and semantic question are supplied to each VLM. Native chat templates and tokenizers remain model-specific. Images are padded to a square before preprocessing; displayed maps are transformed back to the original photograph, including its aspect ratio and bounding-box coordinates. Comparisons within a model hold token positions fixed. Comparisons across models preserve semantic targets, since native subwords and visual grids differ. The generation-steering comparison separately uses COCO validation image 310196 across three VLMs, retaining the original image geometry (Figure 13).

For a fixed answer $a = (a_1, \dots, a_m)$, the shared score is

$$S(x, a) = \sum_{j=1}^{m} \log P(a_j \mid x, a_{<j}), \qquad \Delta S = S(x \text{ with intervention}, a) - S(x, a).$$

The score includes every native answer piece and excludes EOS. A negative $\Delta S$ means reduced support for that answer. Native generation is shown separately so that an explanation of a forced answer is distinguishable from an explanation of the model's actual response. Different tokenizations make absolute scores unsuitable as a cross-model calibration ranking.

The comparison groups also depend on the method's analysis unit. Table 3 outlines these groups. GPT-2 and ViT studies retain the original text and vision settings for tools whose calibrated artifacts or audited computational graphs are not yet available on all five VLMs. These studies complement the shared cases

rather than forming one universal method leaderboard. Training, checkpoint revisions, literal prompts, and validation checks are provided in Appendix A.

The SAE studies extend this design with frozen language and spatial dictionaries, independent calibration and test data, and reconstruction and spatial-shuffle controls. Visual SAE comparisons retain source-native transforms and patch coordinates across CLIP, DINO, SigLIP and Qwen variants. These coordinates are distinct from the square-padded shared-case displays.

The comparisons cover the paper-linked analysis and intervention methods. Attention profiling, head-logit inspection, temperature scaling, and reweighting are also available as configurable utilities; they are not counted as separate paper-reproduction studies.

**Table 3.** Comparison groups and the evidence each provides. Groups organize matched inputs and compatible measures; distinct analysis units remain separately interpreted.

| Family | Methods demonstrated | Matched comparison |
|---|---|---|
| Lens | Logit, Tuned, Jacobian, Attention Lens; EmbedLens readout; Patchscope | Residual/head decoding on IOI; visual-token and transfer readouts on GQA |
| Attention | Rollout, Grad-CAM, ATTATTR, TAM, Beyond-H/C, generic relevance | Seven attribution methods on GQA across five VLMs; native ViT maps and deletion tests |
| Causal | Activation/attribution patching, knockout, path patching, EAP/EAP-IG, VSV steering | Exact versus approximate replacements; attention knockout; controlled sender paths; edge retention; generation changes across three VLMs |
| SAE | ReLU, TopK, Gated, JumpReLU, BatchTopK, Matryoshka; RA-SAE and SAE++ readouts | Frozen Pythia dictionaries; source-native spatial readouts across nine vision/VLM backbones; image–question attribution and feature removal in five decoder configurations |

# 3 Lens methods

## 3.1 Residual and head readouts expose different levels of a decision

Logit Lens applies the model's final normalization and unembedding to an intermediate residual (nostalgebraist 2020). Tuned Lens first applies a fitted affine translator, aligning that representation with the final predictive basis (Belrose et al. 2023). At layer $\ell$ and position $t$,

$$p^{\text{logit}}_{\ell,t} = \text{softmax}(W_U \,\text{Norm}(h_{\ell,t})), \quad p^{\text{tuned}}_{\ell,t} = \text{softmax}(W_U \,\text{Norm}(A_\ell h_{\ell,t} + b_\ell)).$$

Figure 2 shows the same yellow-shirt GQA question across five VLMs. Rows are decoder layers; columns are selected native prompt and answer positions. Cell text gives the most likely vocabulary fragment after native tokenizer decoding, while color gives the rank of the fixed next token. The top-1 search covers the full vocabulary without restricting the language, so English inputs can yield non-English fragments. These are intermediate vocabulary projections, not generated answers or evidence of a particular internal reasoning language. Most context columns are native chat-template suffixes, rather than aligned question words across models. The outlined answer columns expose a concrete disagreement: Qwen3-VL ranks Yellow second at the final readout and generates Black; the other models generate Yellow. Early decoded fragments describe an evolving vocabulary projection, but cannot by themselves establish that the shirt was recognized or locate a causal loss of visual information.

Two other lenses change the readout question. Jacobian Lens uses a calibrated causal transport to the final residual basis to expose verbalizable content (Gurnee et al. 2026). Attention Lens learns a vocabulary decoder for an individual head's projected output (Sakarvadia et al. 2023). These are different units and

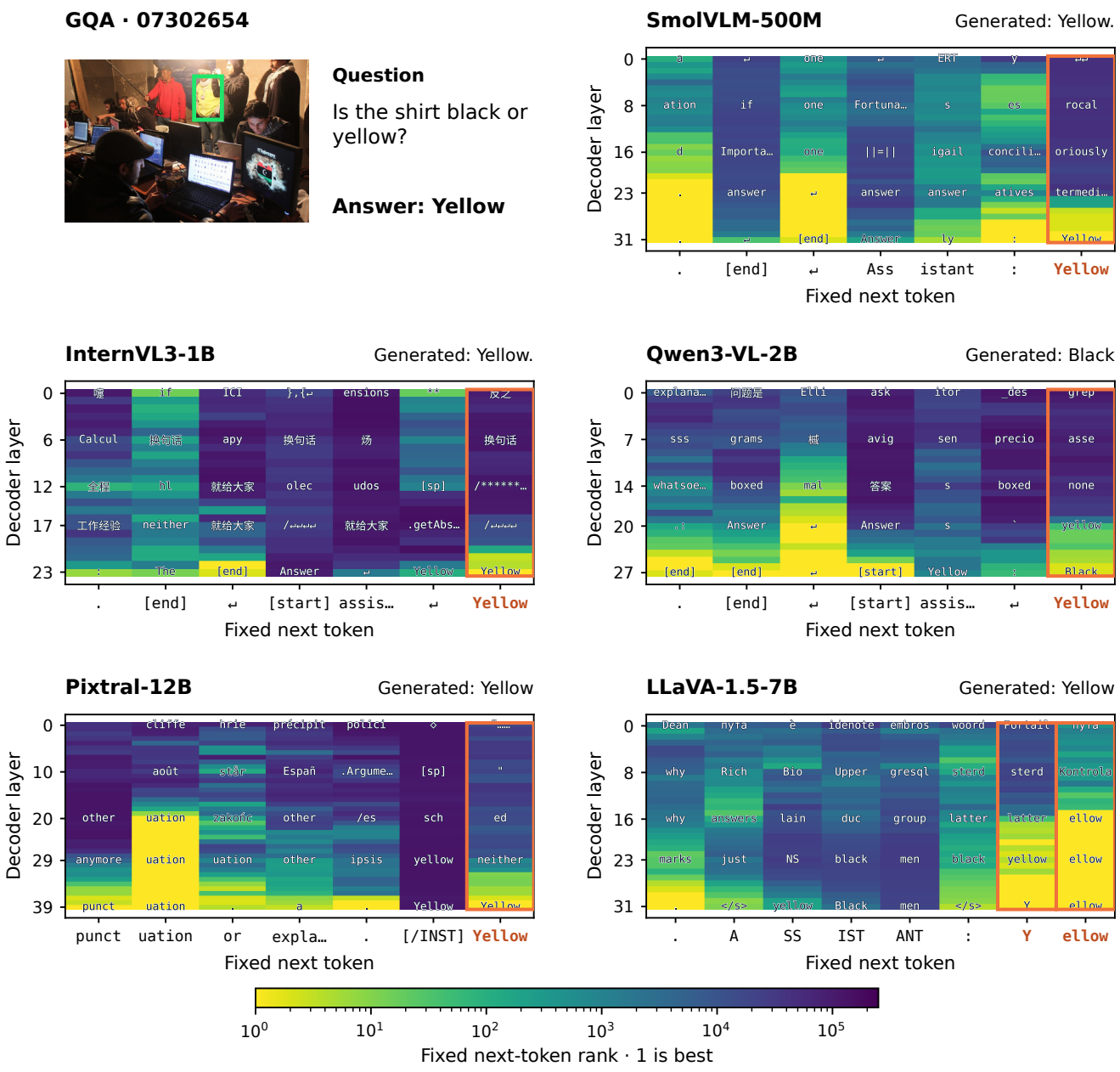



**Figure 2.** Layer-wise vocabulary readouts for one GQA decision across five VLMs. The input panel shows the question, answer and annotated shirt. All decoder layers retain their measured colors; selected rows show shortened top-1 fragments after native tokenizer decoding. Top-1 search includes all vocabulary languages, regardless of the English input. Undecodable or unrenderable fragments remain unlabeled. Yellow indicates a better rank for the fixed next token, using one shared logarithmic scale. Orange outlines identify answer predictions: Qwen3-VL generates Black while the other models generate Yellow. LLaVA predicts Yellow in two pieces, with the second conditioned on the first. Horizontal labels shorten chat markers to [start]/[end], omit whitespace markers and display line breaks as arrows; columns follow each model's native template.

objectives: head readouts resolve components combined in a residual, whereas Jacobian readouts need not represent the next-token predictive distribution.

We examine all four lenses on a GPT-2 IOI prompt ending "Andrew gave a computer to," whose correct next name is Danielle. The native next-token probabilities are 0.842 for Danielle and 0.003 for Andrew. Figure 3 holds the final prediction position and target token fixed. Logit and Tuned Lens first select Danielle at layer 9. At layer 7, calibration improves her rank from 1,880 to 897, while Jacobian Lens ranks her 457th and instead decodes "kids." At layer 8, Jacobian Lens places Danielle at rank 35, compared with 482 for Logit Lens and 365 for Tuned Lens. All three select Danielle from layer 9. Attention Lens reveals variation within layer 10: heads 0, 2, 6, and 10 decode Danielle, while the other eight do not.




**Figure 3.** Residual and head readouts of the same IOI decision in GPT-2 small. Left: Logit, Tuned, and Jacobian Lens track Danielle at the final prediction position. Right: Attention Lens distinguishes heads within layers 7 and 10. Top-token labels and target ranks reveal agreement at late residual sites alongside substantial head-level variation.

This distinction is useful when selecting subsequent interventions. A residual lens traces how a prediction emerges; a head lens nominates finer components for testing. A decoded name alone does not establish

that a head is necessary or identify its functional role. Comparing calibrated head and residual probabilities as if they were additive causal contributions would obscure this difference.

Tuned Lens is calibrated against the final next-token distribution, whereas Jacobian Lens estimates downstream causal transport and Attention Lens decodes individual head messages. Those objectives do not define one shared fidelity metric. We therefore compare their visible readouts on matched inputs, rather than treating small-case distribution divergences as a method ranking. The separately fitted VLM translators and their independent validation settings are retained in Appendix A.

### 3.2 Lexical neighbors reveal associations rather than object labels

The implemented EmbedLens readout examines projected visual-token embeddings before the decoder (Fan et al. 2026). For visual token $v_i$ and vocabulary embedding $e_w$, it reports neighbors ranked by $\cos(v_i, e_w) = v_i^\top e_w / (\|v_i\| \|e_w\|)$, with token norms available in the saved outputs. Figure 4 applies this readout to both GQA shirt images. The image annotations attach decoded lexical fragments to the selected visual patches.

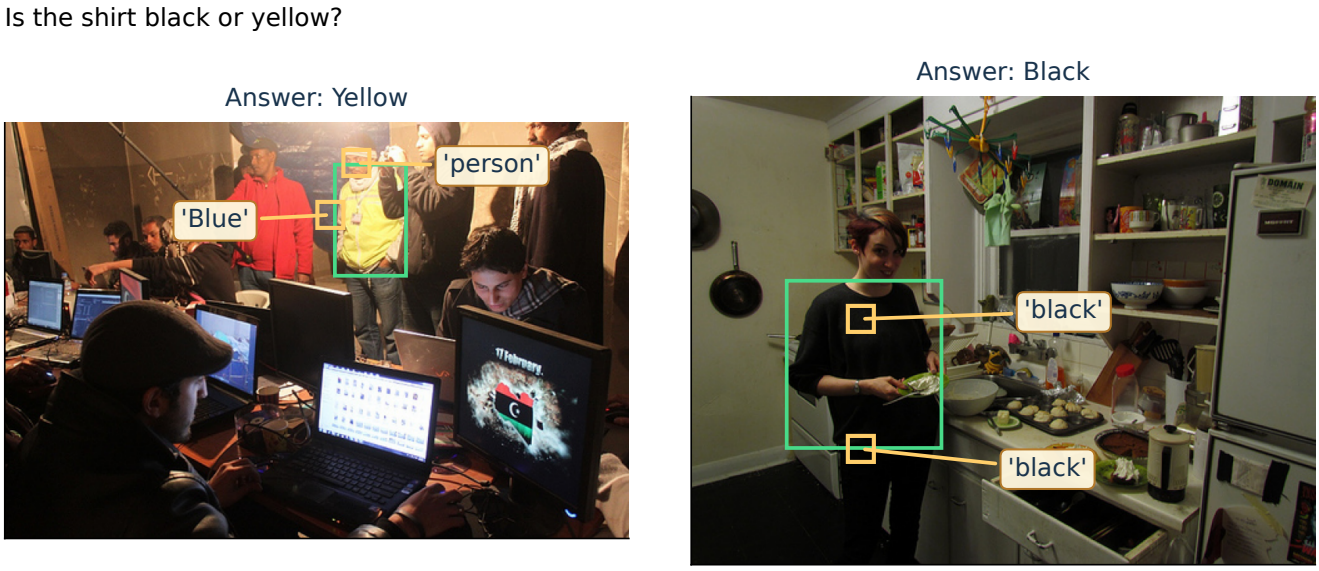



**Figure 4.** EmbedLens attaches lexical readouts to visual patches in LLaVA-1.5-7B. Amber patch boxes and leader lines connect the nearest vocabulary fragments to their projected visual tokens; green boxes mark the annotated shirt. These are embedding-space neighbors rather than independently detected object labels. The black-shirt example associates with "black," while yellow-shirt patches also yield unrelated fragments such as "Blue."

A black-shirt patch associates with "black," while a yellow-shirt patch instead associates with "Blue." Successful and ambiguous neighbors therefore coexist under the same readout. This tool exposes associations cheaply and spatially, but neither cosine similarity nor a large norm is a calibrated object prediction. The library currently implements this semantic readout; the paper's complete sink/dead/alive classification and pruning analysis are outside the demonstrated implementation.

### 3.3 Transfer readouts depend on the decoding prompt

Patchscopes moves a source representation into a specified target computation to make its content readable (Ghandeharioun et al. 2024). We use an identity mapping in LLaVA: an answer-position residual from a GQA source replaces a placeholder residual in a color-copy prompt. Figure 5 shows the source images and word-likelihood changes under a fixed target decoding context, relative to a zero-vector injection at that same target site.

The yellow source at layer 15 increases the complete-word Yellow log probability by 0.419 relative to zero injection, yet its top prediction is "is." Both sources produce "is" at layers 7, 15, and 23, and "->" at layer

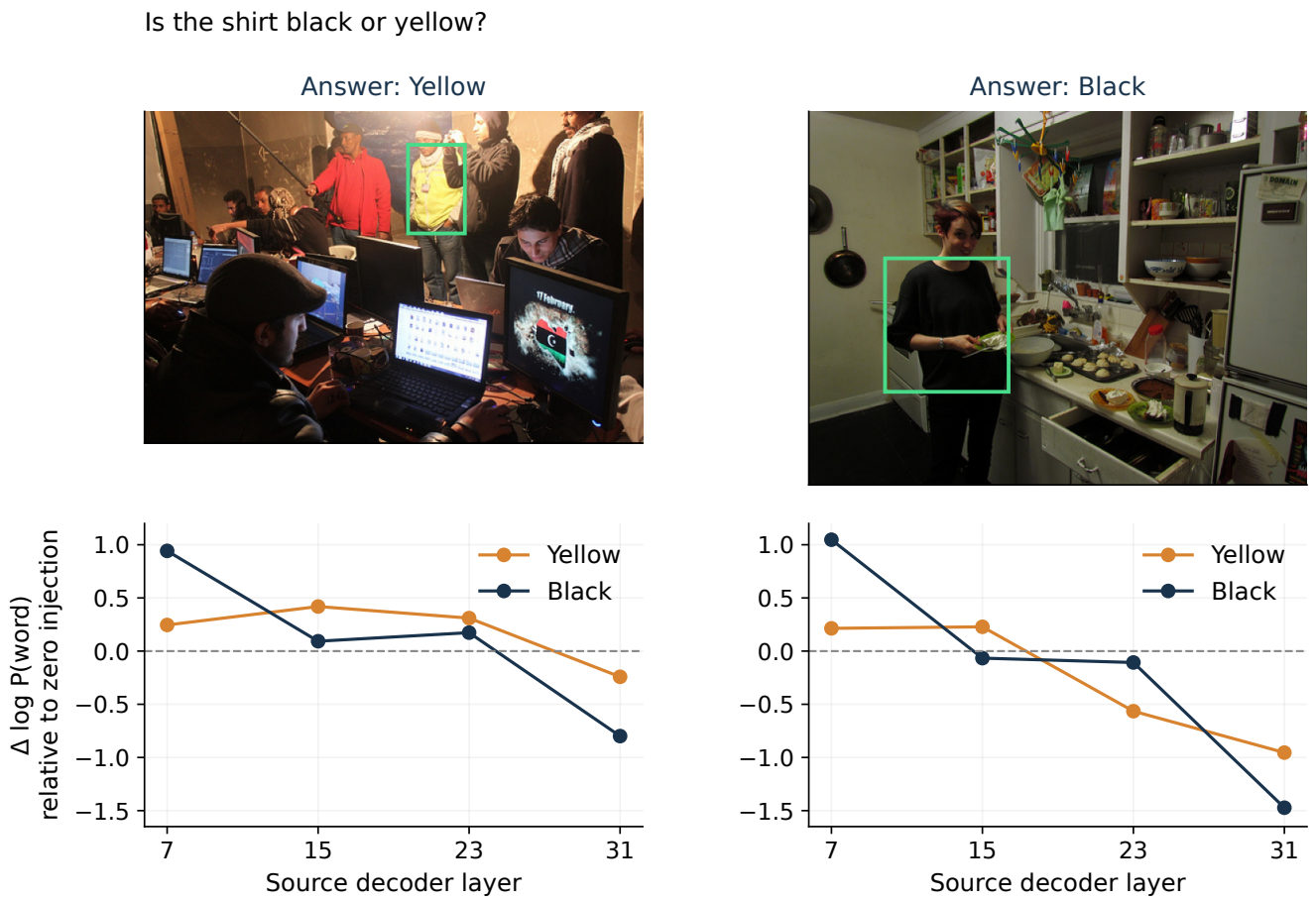



**Figure 5.** Patchscope reads a source residual through a new text context. In LLaVA-1.5-7B, the source answer-position residual replaces the placeholder after target layer 0 in a four-line color-copy prompt, `red -> red`, `blue -> blue`, `green -> green`, then `x ->`. Curves show changes in complete-word log probability relative to a same-site zero-vector injection; positive values indicate greater support for the plotted color. This configuration changes word likelihoods but fails to recover the shirt color as its top continuation.

31. This configuration fails to verbalize the color even though the source computation answers correctly. Unlike direct projection, a transfer readout can use a new decoding context, but its source site, mapping, and target prompt need validation. Likelihood changes alone do not establish successful information transfer.

# 4 Attention explanations

## 4.1 Propagation rules change the map for a fixed answer

Figure 6 compares raw attention and seven explanation methods on the same yellow-shirt GQA input across five VLMs. Reading across a row isolates the method; reading down a column isolates the model. Target-conditioned maps explain Yellow, including Qwen3-VL, whose native answer is Black.

The seven methods use different attribution and propagation rules: Rollout (Abnar and Zuidema 2020), attention Grad-CAM (Selvaraju et al. 2017), ATTATTR (Hao et al. 2021), TAM (Yuan et al. 2021), Beyond Intuition-H/C (J. Chen et al. 2023), and generic relevance (Chefer et al. 2021). Raw attention provides an additional reference. Rollout follows residual-augmented attention connections; Grad-CAM weights attention with a local target gradient; ATTATTR integrates sensitivity along an attention-probability path. The remaining methods use different gradient, value, or relevance-propagation rules. A common output shape makes their maps comparable visually, but does not make their scores equivalent.

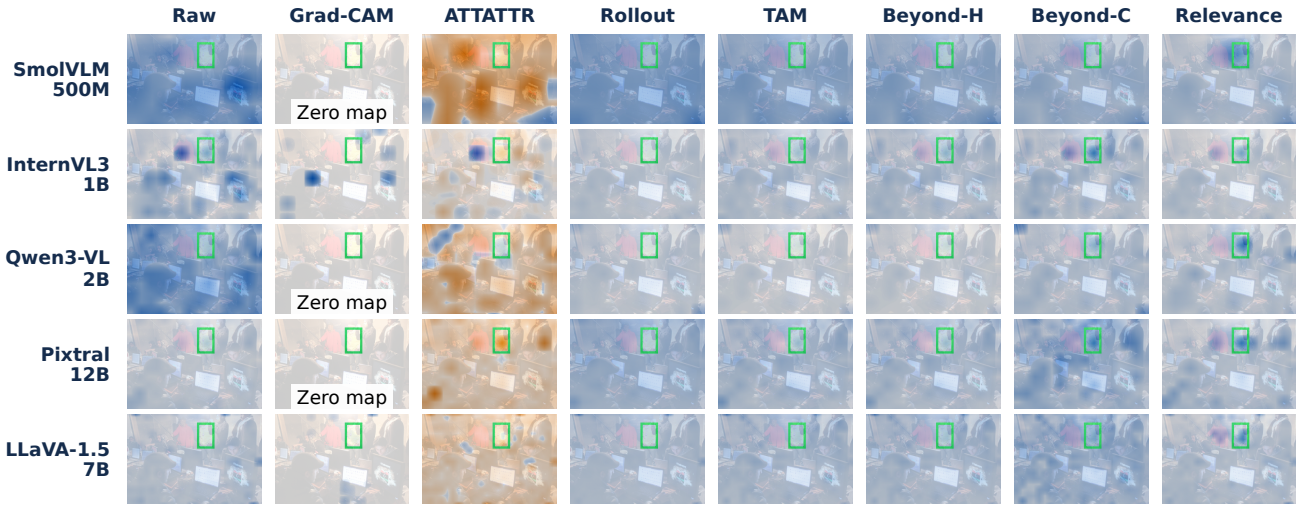



**Figure 6.** Attention explanations assign different spatial emphasis to the same yellow-shirt input. Rows compare five VLMs; columns compare raw attention and seven explanation methods. Target-conditioned methods explain Yellow. Blue indicates positive values and orange negative ATTATTR values; green boxes mark the shirt. Each map is independently normalized, so intensity compares positions within a map rather than score magnitude across maps. Original photographs are displayed at 20% opacity over opaque heatmaps. Raw attention, Grad-CAM, and ATTATTR inspect the last decoder attention layer; propagation methods span the decoder attention stack. "Zero map" denotes a measured null attribution.

The spatial differences are substantial even when the input and target remain fixed. Generic relevance concentrates more strongly within the annotated shirt region than rollout in each displayed model, whereas rollout spreads much of its map outside that object. A sharper map still does not establish a more faithful explanation: the box fractions describe this one input and each map has its own scale.

ATTATTR adds a different distinction. Its signed results are predominantly negative in four models and more mixed in InternVL; discarding the sign would conceal that difference from positive-only maps.

## 4.2 A zero local map can coexist with visual dependence

SmolVLM, Qwen3-VL, and Pixtral have zero last-layer Grad-CAM maps, yet their TAM, Beyond Intuition, and generic relevance maps are nonzero. These methods inspect different paths or aggregate connections under different clipping rules. The actual residual replacement in Section 5.1 also changes SmolVLM's answer score. A zero local, rectified heatmap therefore does not establish that the image is irrelevant to the model. Local Grad-CAM offers a focused and inexpensive view; propagation examines broader dependencies while introducing additional modeling assumptions.

The adapters now expose the attention probabilities actually multiplied by values in the supported eager decoders. ATTATTR therefore uses the same public call across Llama, Qwen2, Qwen3-VL, and Mistral, without an architecture- specific user hook. A measured zero remains a result rather than an unavailable method. Qwen3.5's separate hybrid-stack limitations are discussed in Appendix A.2.

## 4.3 Deletion tests reveal disagreements hidden by plausible heatmaps

To check these methods on their original vision structure, we use pretrained ViT-B/16 and an author-released cat–dog image. This control is a vision classifier, not a VLM column omitted from the preceding comparison. Figure 7 combines the seven maps and their matched deletion curves in one figure. The image is unchanged between rows; the explained class changes from cat to dog. Target-independent rollout consequently stays the same, whereas target-conditioned tools can separate the objects.

Deleting 39 of 196 patches leaves cat probability at 65.1% under rollout, compared with 0.27% under Beyond-C; the original value is 68.6%. Conversely, the dog Grad-CAM map looks recognizable, but its

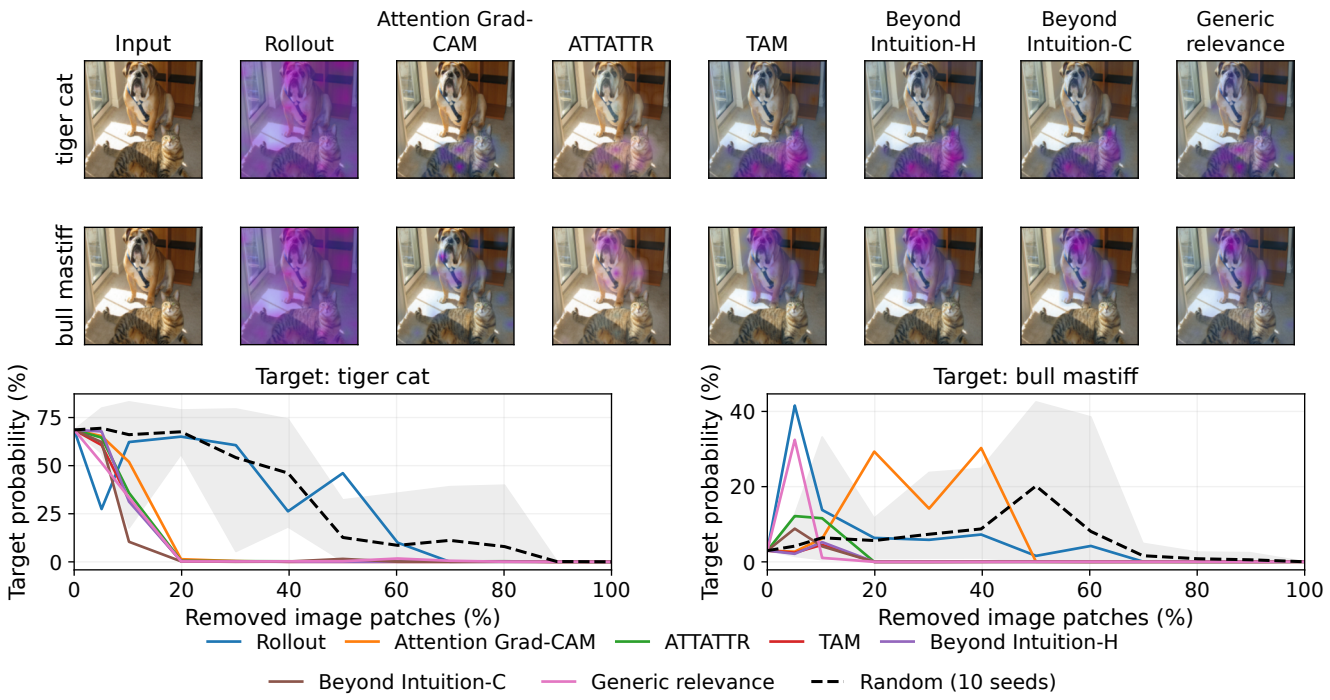



**Figure 7.** Object-conditioned attribution and deletion on ViT-B/16. Seven maps explain cat and dog in the same author-provided image. Curves measure class probability after replacing top-ranked patches with training-mean RGB. Rollout combines target-independent connections, while other tools can separate the classes; deletion exposes differences that map appearance alone does not resolve.

deletion raises dog probability from 2.99% to 29.3%. A visual explanation and the effect of removing its selected regions can thus disagree. Deletion supplies an intervention-based diagnostic for choosing among maps, while remaining sensitive to the replacement baseline and removal of competing objects. The library supports both parts of this comparison rather than treating visual sharpness as sufficient evidence of faithfulness.

# 5 Causal interventions

## 5.1 Attribution patching can miss large replacement effects

Activation patching replaces a selected activation with its value from a reference input and recomputes downstream behavior (Meng et al. 2022). Attribution patching estimates the same replacement from a local gradient (Nanda 2023). For a residual $h_{\ell,S}$ at token set $S$ and aligned reference value $d_{\ell,S}$,

$$\Delta_{\ell,S} = S(x; h_{\ell,S} \leftarrow d_{\ell,S}) - S(x), \qquad \widehat{\Delta}_{\ell,S} = \left\langle \nabla_{h_{\ell,S}} S(x),\ d_{\ell,S} - h_{\ell,S} \right\rangle .$$

We compare these effects on the same GQA input in SmolVLM and LLaVA. The reference input is a gray image with identical dimensions, prompt, and answer prefixes. Figure 8 distinguishes replacement of all visual states from replacement of the answer-predicting states. These are hidden-state interventions: the image and text of the patched run stay fixed. Answer-predicting states are the residual vectors used to predict each piece of the scored answer; a multi-piece answer can involve more than one position. Actual effects are measured at seven layers; first-order estimates cover every layer. The diagram shows the two alternatives, tested in separate runs.

SmolVLM's layer-5 visual replacement changes $\log P(\text{Yellow})$ by $-0.498$, compared with an estimate of $-0.067$. LLaVA's corresponding effect is $-1.646$, while the estimate is $+0.100$, reversing the sign. Large blank-image replacements leave the neighborhood described by a local gradient. Attribution patching

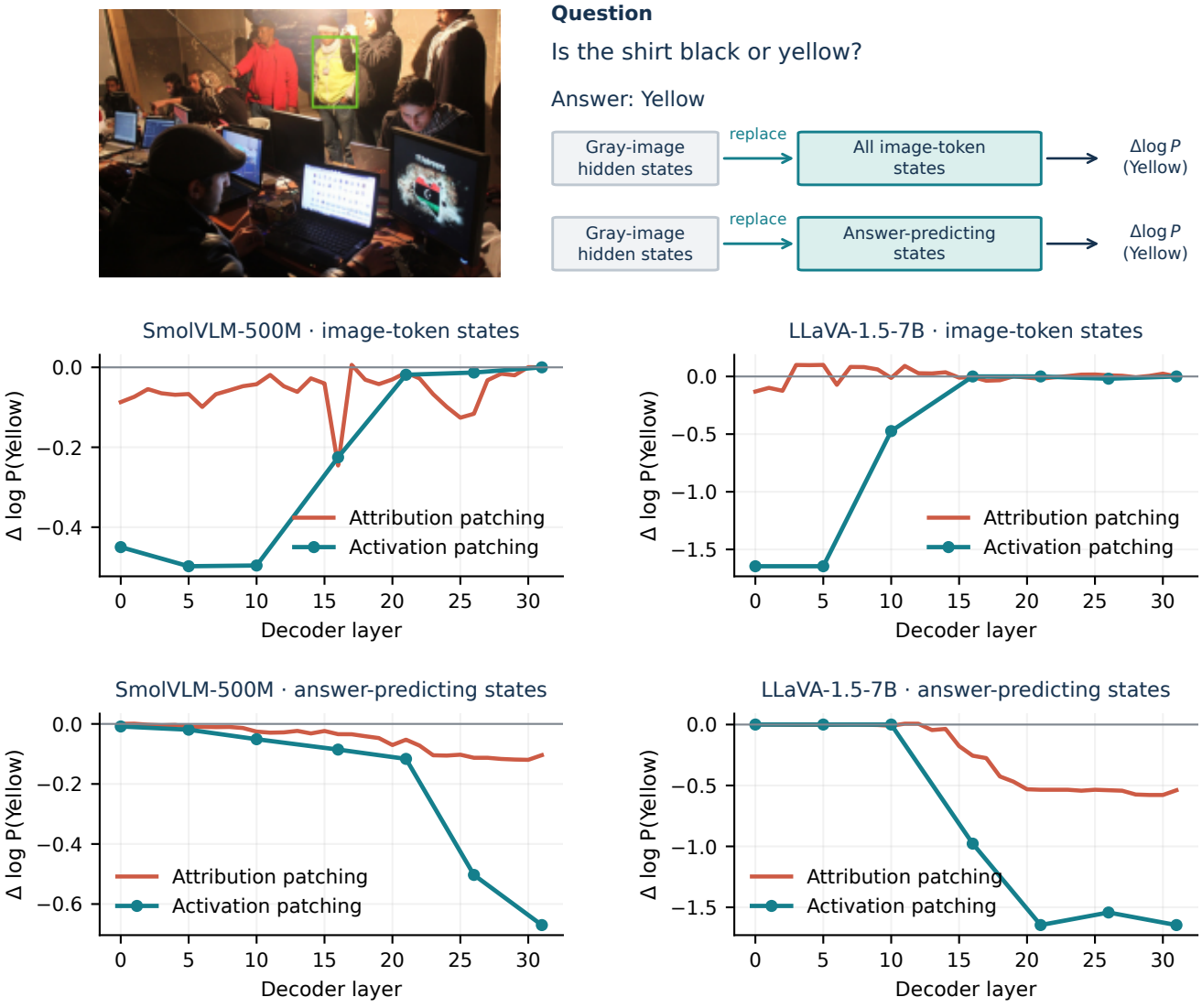



**Figure 8.** Attribution patching can miss effects measured by activation patching. Gray-image hidden states replace selected states in the real-image run at one decoder layer. The schematic rows are separate interventions: all image-token states or the states that predict the answer's token pieces. Teal markers measure changes in the complete-answer log probability; red curves estimate the same changes from gradients. Negative values reduce support for Yellow. The estimates miss large early image-state and late answer-state effects.

can cheaply nominate candidate sites, but these results motivate testing important candidates with actual replacement. They do not imply that every smaller head- or token-level estimate will fail.

## 5.2 The intervention unit determines the apparent locus of dependence

Image-slot replacement is strongest early in the yellow-shirt case; replacing prediction slots has larger late effects. The final visual-slot effect is zero because no remaining decoder layer transports the change into an answer position. That zero is a consequence of the intervention location, rather than evidence that the model ignored the image.

Figure 9 refines the experiment to one native text position at a time. Rows are layers and columns are question or assistant-boundary subwords; actual and estimated effects share a signed scale. Replacing the question word "yellow" at an intermediate layer changes the answer score in both models. The largest late effects occur at the final prediction position, whose role makes this sensitivity expected. This distinguishes sensitivity at question-token sites from disruption of the answer readout itself.

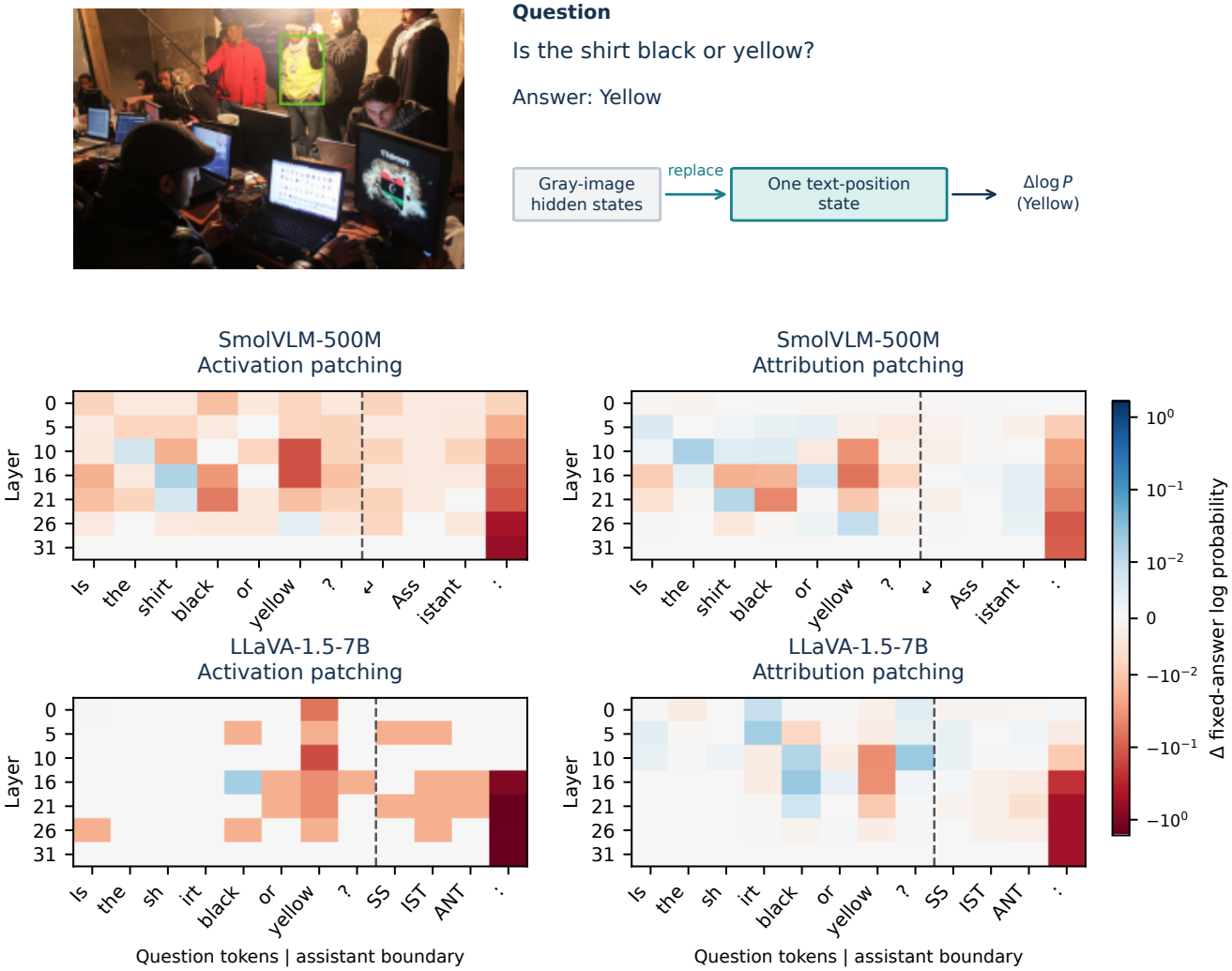



**Figure 9.** Single-position replacement separates question-word effects from prediction-site sensitivity. Each cell replaces one native text-position state with its corresponding gray-image state at the indicated layer, as depicted in the schematic. Columns cover question and assistant-boundary subwords. Red reduces the complete-answer log probability of Yellow; blue increases it. Activation patching measures the replacement effect and attribution patching estimates it, using the same signed scale. Contrast with the joint replacements in Figure 8.

The black-shirt input provides a useful contrast (Figures 20 and 21). LLaVA's largest prediction-slot effect is only $-0.172$, compared with $-1.646$ on Yellow. The maps therefore characterize a specified input and corruption. Neither two examples nor a visually similar layer pattern establishes a universal pathway or model-level robustness ranking.

## 5.3 Knockout tests information routes, while path patching isolates senders

Attention knockout blocks selected key-to-query connections before softmax, allowing downstream computation to recompute (Z. Zhang et al. 2025). On the yellow-shirt GQA input, we compare image-to-question, image-to-prediction, and question-to-prediction routes over windows of up to nine layers. Figure 10 plots the fixed first-answer-token probability and compares target-region and other-image keys. The displayed question identifies the input. This paper-aligned demonstration uses a single-word-or-phrase instruction and a first-subword metric, rather than the shared complete-answer score; Appendix A records the full instruction.

The image-to-question route is more disruptive than the direct image-to-prediction route on this example. Other-region blockade can instead increase answer support. Knockout tests dependence on a route with-

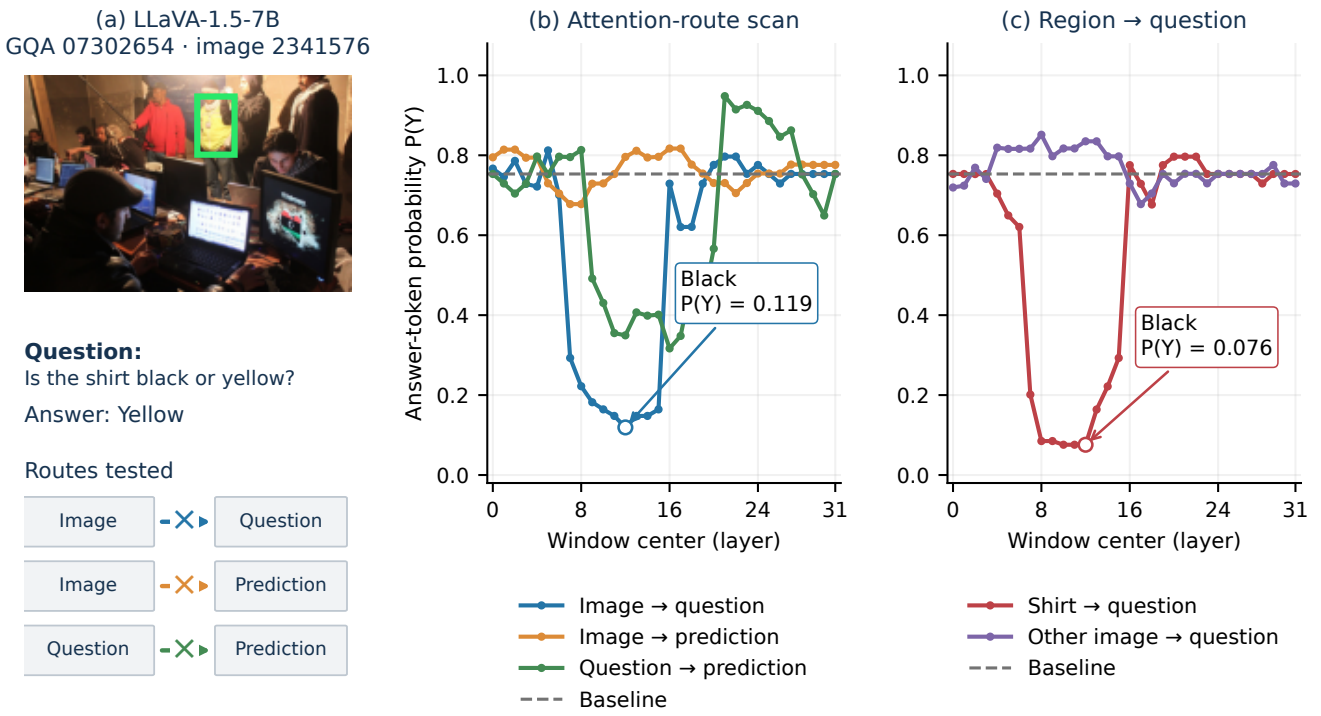



**Figure 10.** Attention Knockout reveals route-specific dependence in LLaVA-1.5-7B. Blocking image-to-question connections at layers 8–16 lowers the fixed Yellow first-subword probability from 0.753 to 0.119 and makes Black the next-token prediction. Blocking shirt keys lowers it to 0.076; region sizes differ, so these curves describe the selected masks rather than controlled localization accuracy.

out requiring a reference input, but changes attention normalization and depends on the mask/window. The target region contains 20 keys versus 556 outside it, so a size-matched control would be required to attribute the difference specifically to grounding.

IOI-style path patching asks a narrower question (Wang et al. 2023). A controlled base run replaces a sender head with its reference-input output while freezing other attention-head outputs to base values. MLPs and normalization recompute; a fresh base run then replays only the resulting receiver activation. Figure 11 sweeps sender heads on a paired MMVP wing-status input in five VLMs, using the final prediction-position residual as receiver.

The largest negative effects include Qwen3-VL L23.H6 ($-0.054$), Pixtral L34.H12 ($-0.080$), and LLaVA L14.H3 ($-0.064$). The heatmaps identify influential senders under this receiver and freezing rule. They are not an automatically discovered complete circuit or functional labels such as "name mover." Path patching offers tighter control of selected paths than unrestricted residual replacement, at the cost of more forwards and a receiver-dependent question.

## 5.4 Circuit attribution requires measured recovery

EAP and EAP-IG rank independently replaceable connections in an explicit computation graph; EAP-IG integrates gradients along the input-embedding path (Hanna et al. 2024). We retain the paper-aligned GPT-2 IOI graph, rather than substituting VLM residual nodes for unimplemented edges. Figure 12 evaluates both rankings by actual edge-retention interventions on the same 32,491-edge graph. Excluded edges receive reference messages, while retained edges remain live. Recovery is measured as

$$R(E) = \frac{m(E) - m(\text{reference})}{m(\text{clean}) - m(\text{reference})},$$

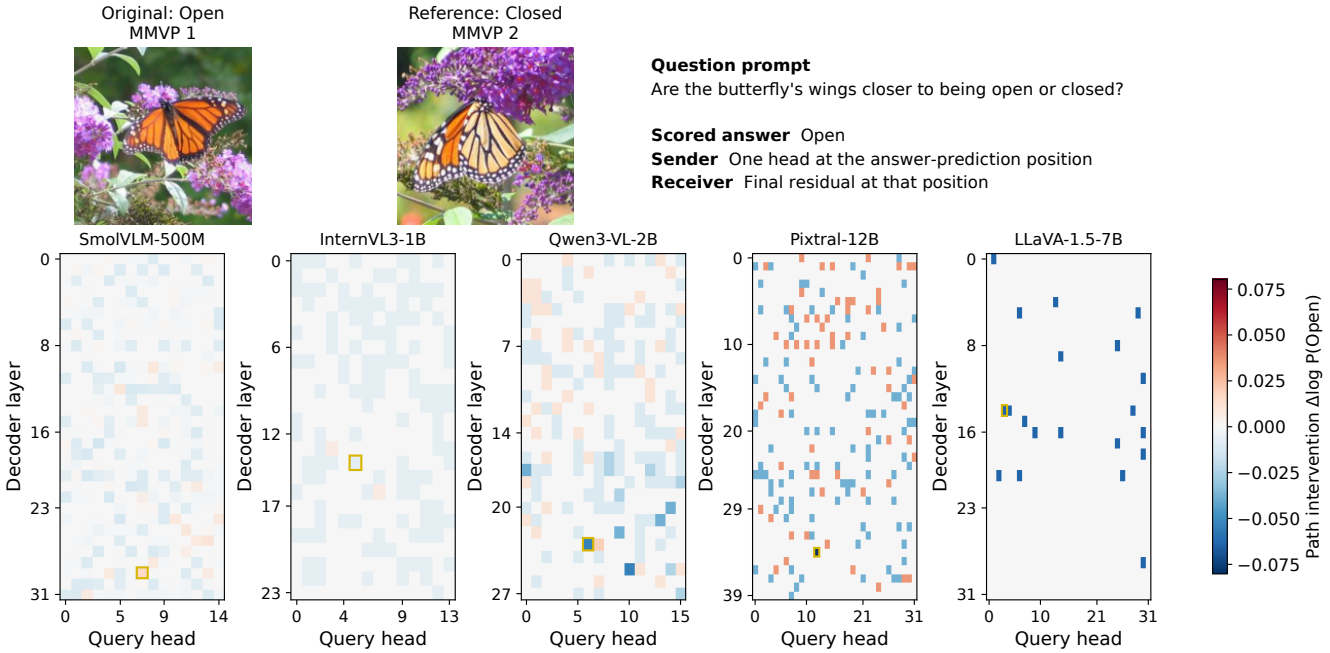



**Figure 11.** Controlled sender-head paths on a paired butterfly-wing input. Each cell replaces one sender using its reference-image value at the prediction position, with other head outputs frozen and only the recomputed final residual replayed. Other head outputs are frozen during the controlled run; MLP and normalization recompute, and only the resulting receiver is injected in a fresh run. Signed effects compare the same configured receiver across five VLMs. Yellow outlines identify the largest absolute effect within each model; blue decreases and red increases support for Open.

where $m$ is the Danielle-minus-Andrew logit margin. This quantity can fall below zero or exceed one; it is not a probability or a sum of attribution scores.

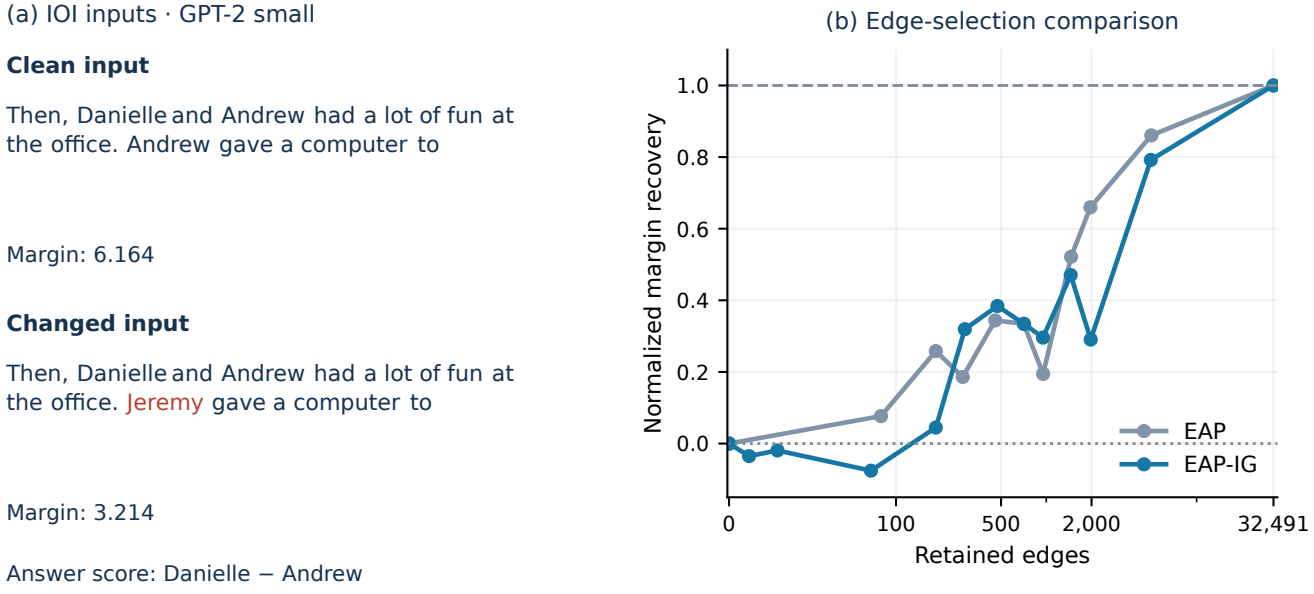



**Figure 12.** Attribution rankings differ in actual IOI circuit recovery. EAP and input-path EAP-IG select connections in the same GPT-2 graph; each curve point is a fresh retained-edge intervention. Both complete graphs recover the clean margin, but EAP-IG does not uniformly outperform EAP on this single prompt pair.

At 184 retained edges, EAP recovers 0.258 of the original-reference margin difference, while EAP-IG recovers 0.045. Integrating gradients therefore does not guarantee a better selected circuit on each instance. The practical benefit of this comparison is the separation of ranking and validation: a high attribution score proposes a connection, and edge retention tests the proposed subgraph. The graph-specific intervention contract and single-pair evaluation limit what can be inferred about general circuit quality.

## 5.5 Steering depends on both model and strength

VSV, the visual steering component of VISTA, adds an image-derived residual direction during generation (Li et al. 2025). Figure 13 holds a COCO image and its description request fixed across LLaVA, SmolVLM and Qwen3-VL. Each row compares the model's baseline with two smaller steering scales. Exact excerpts retain the changed content, and ellipses indicate omissions. These outputs examine generation changes directly, rather than necessity under a corruption.

COCO2014 · image 310196



Prompt: Please help me describe the image in detail.

| | Baseline | VSV · lower λ | VSV · higher λ |
|---|---|---|---|
| **LLaVA-1.5 7B** | **λ = 0**<br>… The snowy terrain is dotted with various tracks, indicating that the slope is frequently used by snowboarders and skiers. … | **λ = 0.02**<br>… The snowy landscape is dotted with several snow-covered rocks scattered throughout the scene, adding to the wintry atmosphere. … | **λ = 0.05**<br>… The snow-covered slope is a popular skiing and snowboarding spot, with a few other people scattered around the area, likely enjoying the winter sports. … |
| **SmolVLM 500M** | **λ = 0**<br>… The snowboard is a standard snowboard, characterized by its flat, rectangular shape and a single, large, flat, circular nose. … | **λ = 0.02**<br>… The individual is captured mid-air, … They are wearing a red and black outfit, … The person is holding the snowboard with both hands, … | **λ = 0.05**<br>… The individual is captured mid-air, … They are wearing a red and black outfit, … performing a snowboard trick. … |
| **Qwen3-VL 2B** | **λ = 0**<br>… The central focus is a single skier, seen from above, who is carving down the slope. … | **λ = 0.02**<br>… The skier, positioned in the lower-right quadrant of the image, is the main focal point. … | **λ = 0.05**<br>… A single person is the focal point of the image, positioned in the lower-right quadrant, actively skiing down the slope. … |

**Figure 13.** The same image-description request under residual VSV in three models. Rows hold the model fixed; columns compare its baseline with two steering strengths, printed in each cell. Excerpts preserve the generated wording, with ellipses marking omitted text. Green highlights newly mentioned details supported by the image; red highlights selected claims without image support. Uncolored text is not fully audited. The common intervention follows the raw residual rule, without SLA.

LLaVA illustrates why fluency and grounding need separate inspection. Its baseline already mentions tracks but infers frequent use of the slope. At strength 0.02, it instead mentions visible rocks; at 0.05, it invents other people. All three descriptions are grammatical. The retained 0.17 stress run repeats snow-related text even after exact text-token alignment.

SmolVLM changes from a detailed equipment description to a maneuver description, adding the visible red-and-black clothing but also an unsupported airborne pose and board-holding claim. Qwen3-VL retains a single-rider account while changing its spatial framing and track description. The same raw scale therefore changes different claims in different models; added visual detail and newly invented content can occur within the same description.

Our intervention uses the paper's raw block-residual rule, without SLA. The relative additive displacement depends on $\lambda\|v_\ell\|/\|h_\ell\|$, so identical scales need not produce equivalent perturbations across models. The released VISTA code instead uses normalized MLP-output directions and additional direction processing. Its numerical scale cannot be assumed to transfer to this implementation.

Figure 14 tests local evidence under fixed prefixes rather than comparing the changing prefixes of those descriptions. The maps give person/car token ranks at the same 80 baseline prefixes; the bottom curve holds one prefix and prediction position fixed across strengths.

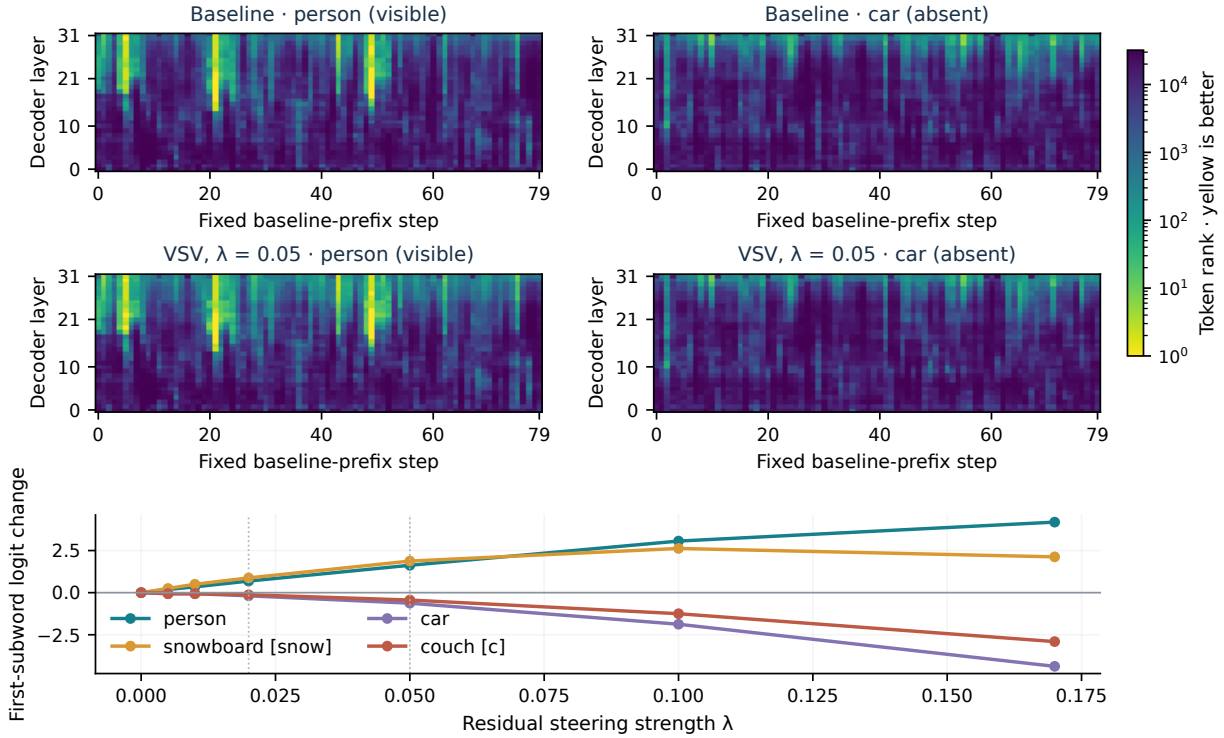



**Figure 14.** Object-token support under fixed prefixes does not guarantee a grounded description. In the matched LLaVA run, baseline and steered person/car heatmaps share the same 80 baseline prefixes and logarithmic rank scale; brighter colors indicate better ranks. The strength curves fix the prefix "The image captures a snowboarder skillfully riding" and measure first-subword logit changes, not complete-word probabilities. At the highest retained strength, favorable person/car responses coexist with repetitive free generation.

At strength 0.17, person's logit increases by 4.19 and car's decreases by 4.38, despite the repetitive free generation. A favorable object-token response therefore need not yield a grounded, fluent description. VSV offers a direct way to alter visual influence, but its strength requires evaluation against both grounding and decoding quality. This is a Steering/VSV study; the SLA component is absent, so it does not represent complete VISTA reproduction.

Across these intervention groups, the library keeps the change and its question explicit. Residual replacement tests a counterfactual activation; knockout tests a blocked route; path patching controls sender influence; edge retention evaluates a proposed circuit; steering changes generation. Matched comparisons clarify differences within those questions, while separate studies preserve the meaning of methods with incompatible interventions.

# 6 Sparse features connect reconstruction, concepts and spatial structure

A sparse autoencoder (SAE) represents a native activation $h$ using an expanded dictionary of features,

$$z = E(h), \qquad \hat{h} = D(z).$$

The encoder selects a sparse feature vector; the decoder reconstructs the original representation. This provides feature coordinates for concept readout and spatial analysis. Feature indices acquire interpretations through measurements on independent inputs. They are not predefined concept labels.

The matched language experiment evaluates six pretrained families. ReLU with an L1 penalty encourages sparsity by penalizing rectified activations (Bricken et al. 2023); TopK retains the largest positive activations at each token (Gao, Tour, et al. 2024). Gated SAEs separate feature selection from magnitude estimation (Rajamanoharan, Conmy, et al. 2024), while JumpReLU learns activation thresholds (Rajamanoharan, Lieberum, et al. 2024). BatchTopK allocates a sparsity budget across training examples and uses a saved threshold for inference (Bussmann, Leask, et al. 2024). Matryoshka BatchTopK additionally trains nested dictionary prefixes to reconstruct activations at several feature budgets (Bussmann, Nabeshima, et al. 2025). Our experiments use published weights throughout: neither the backbone nor any SAE is trained or recalibrated. The same native interface supplies features and reconstruction to the shared evaluation tools.

## 6.1 Matched language dictionaries expose separate trade-offs

The language comparison uses 18 SAEBench dictionaries: three source settings for each family, all attached to Pythia-160M block 8, with input width 768 and dictionary width 4,096 (Karvonen et al. 2025). All checkpoints receive the same 524,288 activations from 512 separate, 1,024-token Pile test-file contexts (Gao, Biderman, et al. 2021). An independent 128-context calibration subset supplies a fixed reference mean $\mu_{\text{cal}}$. We report global reconstruction and realized activity as

$$\text{FVU}_{\mu_{\text{cal}}} = \frac{\sum_i \|h_i - D(E(h_i))\|_2^2}{\sum_i \|h_i - \mu_{\text{cal}}\|_2^2}, \qquad \overline{L}_0 = \frac{1}{N}\sum_i \sum_j \mathbf{1}[z_{ij} \neq 0].$$

The ratio is computed after aggregating all selected positions. Averaging per-batch FVU would change the metric with batch partitioning. Behavioral fidelity is measured on 64 fixed test contexts, containing 65,472 shifted next-token predictions: the SAE reconstruction replaces the selected native residual, and the remaining model recomputes. We record the increase in negative log-likelihood ($\Delta$NLL), distributional KL and loss recovered relative to a separate zero-residual control.

Concept readout uses AG News with 4,000 training, 1,000 development and 1,000 official-test documents (X. Zhang et al. 2015). Nonpadding token features are averaged within each document, using at most 128 tokens. For each one-versus-rest class, the probe selects one, two or five features using training labels only. Training statistics define standardization; development AUROC selects logistic-regression regularization. Raw residual neurons, training-fitted PCA and a fixed random projection use the same splits and feature budgets. Figure 15 places concept readout beside reconstruction and model fidelity. Its horizontal axis is realized Pile activity, rather than a nominal hyperparameter shared across incompatible sparsity objectives.

At approximately 80 active features per token, BatchTopK gives FVU 0.0648 and $\Delta\text{NLL} = 0.1945$ nats, while the ReLU checkpoint at $\overline{L}_0 = 77.71$ gives 0.1078 and 0.5796 nats. The paired NLL-increase difference is $-0.3851$ nats, with a 95% document-bootstrap interval of $[-0.4192, -0.3569]$. TopK at 320 active features reaches FVU 0.0182 and an NLL increase of 0.0341 nats. These results show how activation reconstruction and downstream predictions change along the available sparsity settings; L1 penalties do not impose the same activity counts as TopK budgets.

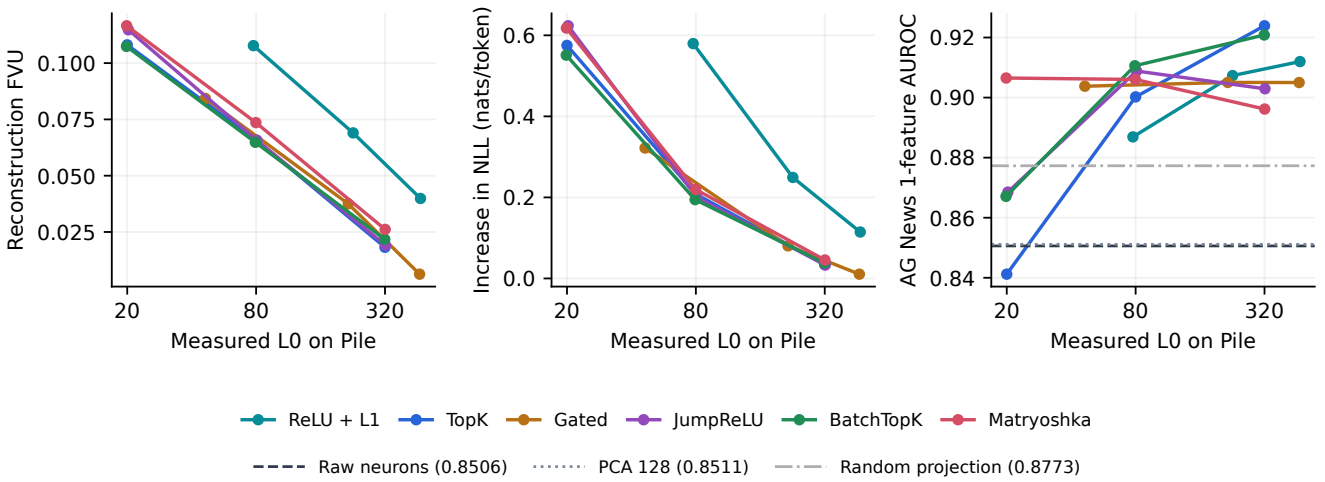



**Figure 15.** Pretrained language SAE trade-offs under a common backbone, layer and dictionary width. Each point is one of 18 fixed checkpoints. Left: Pile reconstruction FVU with an independent calibration mean. Middle: next-token NLL increase after native residual reconstruction. Right: held-out AG News macro AUROC using one selected feature per one-versus-rest class; horizontal lines show matched raw-neuron, PCA and random-projection controls. All panels use measured Pile $\overline{L}_0$ on the horizontal axis. All SAE weights remain frozen during evaluation.

Concept readout gives a different ordering. Near 20 active features, Matryoshka's one-feature AUROC is 0.9065 versus TopK's 0.8412, although its FVU is higher, 0.1166 versus 0.1081. Increasing the number of classifier features also changes the comparison: five PCA components per class reach AUROC 0.9481, above every evaluated five-feature SAE probe, whose highest AUROC is 0.9380. A reconstruction advantage therefore does not establish a concept-readout advantage, and a useful sparse feature does not establish that the dictionary is the best representation for every classifier budget. This separation motivates reporting several measured behaviors rather than collapsing SAE quality into one benchmark score (Chanin 2026). Appendix A.6 lists every checkpoint and the probe controls.

## 6.2 Visual features expose spatial structure across models and methods

We ask whether compact SAE features retain useful object-associated spatial structure, and whether good reconstruction also gives reliable localization. Twelve pretrained configurations cover nine backbones and seven recipes (Joseph et al. 2025; Pach et al. 2025; Gohil et al. 2026; Zhao et al. 2026; Fel et al. 2025). ReLU and TopK share block-11 residual observations within CLIP-B/32 and DINO-B/16, providing the closest within-backbone comparisons. The other rows evaluate published configurations at their native sites. Eleven observe visual encoders; Qwen2.5 observes decoder-layer-34 visual tokens. CLIP's Gated SAE uses block-9 MLP outputs, Qwen3 uses vision-layer-23 SAE++ features, and RA-SAE uses DINOv2 with registers. These site and backbone differences are retained when interpreting the comparison.

Every model receives the same fixed 500 COCO2017 training and 1,000 validation source images. A fixed category list and training-only inside-versus-outside box contrast, divided by feature RMS, select one feature per category and dictionary before testing. Native transforms can change visible category-box unions and eligible pair counts. Backbone weights, SAEs and inference thresholds remain frozen; Appendix A.6 records their source contracts.

For native spatial representation $h_{u,v}$, the map is $M_j(u,v) = [E(h_{u,v})]_j$, reshaped on the actual visual-token grid. Reconstruction retains the author's full feature vector, including signed coefficients; selection and localization use $M_j^+ = \max(M_j, 0)$. Figure 16 fixes the source photograph, backbone and block-11 residual observation within each ReLU–TopK row.




**Figure 16.** ReLU and TopK SAE features on the same photograph for CLIP-B/32 and DINO-B/16. Rows fix the backbone and observation layer; columns compare dictionaries. Blue heatmaps show feature activation; orange boxes mark people.

Table 4 separates two questions. Full-dictionary $\overline{L}_0$ and FVU measure activity and reconstruction over every native spatial position, using a training-only reference mean for FVU. Localization evaluates the single training-selected feature for each category. Coverage is the fraction of eligible pairs with positive response; Point@1 counts a hit when its strongest location falls inside a category box, treating inactive maps as misses. Mass averages the fraction of positive evidence inside boxes over responding pairs only. Shuffled Point@1 preserves feature values while permuting their spatial locations. These spatial scores are percentages; $N$ counts eligible image–category pairs after native cropping. Separately evaluated CLS readouts remain in Table 12.

**Table 4.** Pretrained spatial SAE configurations on COCO. Activity and reconstruction are evaluated alongside category-feature response and localization; shuffled maps provide a spatial control.

| | | Activity / fidelity | | Spatial response | | | | |
|---|---|---|---|---|---|---|---|---|
| Model | Recipe | $\overline{L}_0$ | FVU↓ | $N$ | Coverage | Point@1 | Shuffled | Mass |
| CLIP B/32 | ReLU | 1,413.14 | 0.0152 | 1,534 | 100.0 | 54.5 | 19.2 | 39.4 |
| CLIP B/32 | TopK | 64.00 | 0.2163 | 1,534 | 66.5 | 44.7 | 16.2 | 58.8 |
| CLIP B/32 | Gated | 107.80 | 0.1261 | 1,534 | 69.4 | 47.1 | 16.3 | 57.8 |
| DINO B/16 | ReLU | 1,083.63 | 0.0063 | 1,534 | 97.9 | 52.7 | 18.6 | 38.6 |
| DINO B/16 | TopK | 1,023.82 | 0.0085 | 1,534 | 65.4 | 35.5 | 15.6 | 44.4 |
| CLIP L/14-336 | Matryoshka | 23.09 | 0.1378 | 1,534 | 83.1 | 62.5 | 18.5 | 64.9 |
| CLIP B/16 | BatchTopK | 17.05 | 0.3525 | 1,534 | 52.7 | 41.1 | 15.1 | 70.8 |
| SigLIP B/16 | BatchTopK | 252.68 | 0.2634 | 1,636 | 90.6 | 51.8 | 17.0 | 44.8 |
| DINOv2 B/14 | BatchTopK | 995.02 | 6.6730 | 1,471 | 98.2 | 66.8 | 23.8 | 52.3 |
| Qwen2.5-VL-3B decoder L34 | TopK | 100.00 | 0.2078 | 1,636 | 70.7 | 45.2 | 15.0 | 53.6 |
| Qwen3-VL-4B vision L23 | SAE++ L1 | 6.21 | 0.0087 | 1,636 | 86.3 | 55.5 | 16.1 | 54.3 |
| DINOv2 B/14 + registers | RA-SAE | 10.30 | 0.2977 | 1,471 | 36.4 | 29.0 | 9.9 | 70.6 |

The matched CLIP-B/32 observations show a clear compactness–coverage trade-off. TopK uses 64 active features per position versus ReLU's 1,413, but FVU rises from 0.0152 to 0.2163 and Point@1 falls from 54.5% to 44.7%. Its greater active-map mass inside boxes (58.8% versus 39.4%) coexists with lower response coverage (66.5% versus 100%). For these configurations, TopK produces more concentrated responses when active; ReLU reconstructs better and responds more reliably across image–category pairs.

The DINO-B/16 pair has similar activity (1,084 versus 1,024 features), yet ReLU reaches 52.7% Point@1 versus TopK's 35.5%. Their pointing rates conditional on response are nearly equal (53.8% and 54.3%); the overall difference mainly follows coverage, 97.9% versus 65.4%. Thus an active-only localization score can hide missed responses, even without a large sparsity difference. Gated's distinct MLP hook prevents isolating a gating effect from its row.

Compact spatial readouts are nevertheless feasible: CLIP-L/14 Matryoshka uses 23.09 active features, with 83.1% coverage and 62.5% Point@1. Across the broader configurations, reconstruction and localization need not agree. DINOv2 BatchTopK has FVU 6.6730, worse than the fixed calibration-mean baseline, but reaches 66.8% Point@1 versus 23.8% shuffled. RA-SAE's 70.6% active-map mass likewise accompanies only 36.4% coverage and 29.0% overall Point@1. These results support choosing configurations using reconstruction, coverage and localization together; neither low FVU nor concentrated responding maps establish reliable object-associated features on their own.

Figure 17 compares four fixed photographs across six visual-encoder backbones, including DINOv2 variants with and without registers. Its first column gives uncropped originals; each remaining column uses one fixed dictionary and the backbone's native image transform. All columns observe visual representations before language decoding.

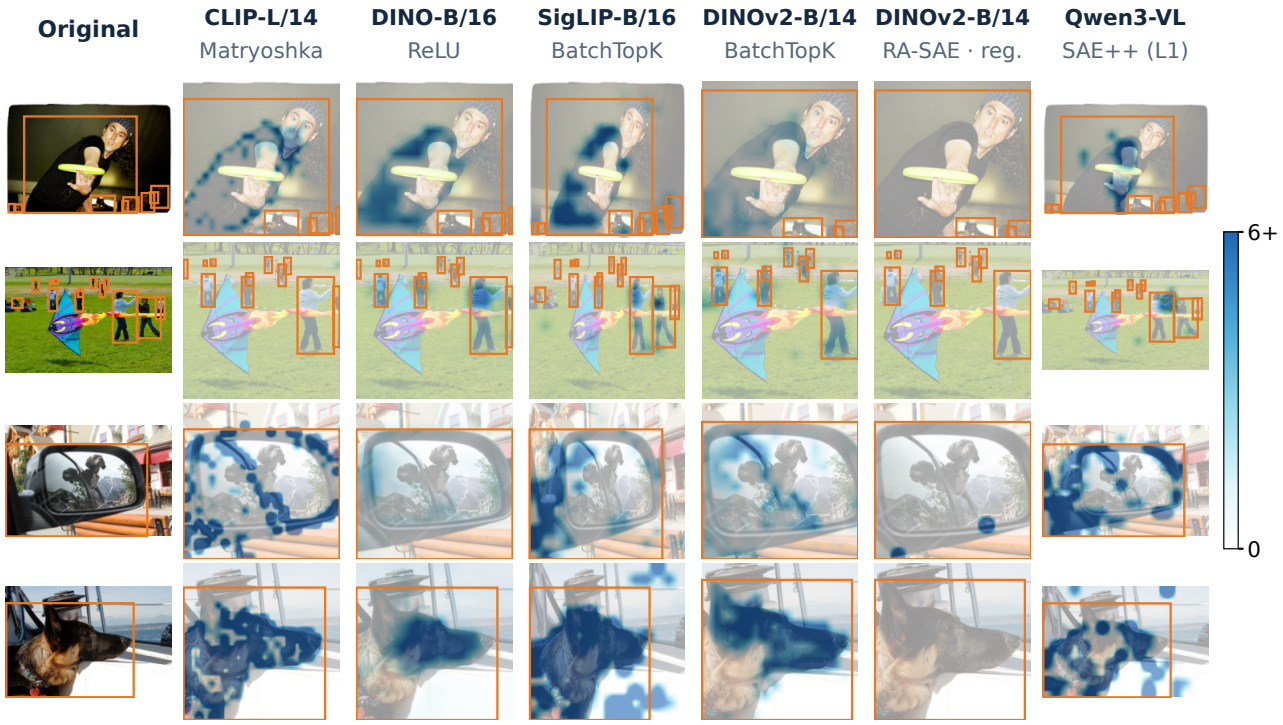



**Figure 17.** Visual-encoder SAE features across six backbones on four photographs. Each column uses one pretrained dictionary at its native observation layer. Blue heatmaps show feature activation; orange boxes mark target objects. All SAE observations precede language decoding.

All twelve configurations exceed their spatially shuffled controls, but coverage ranges from 36.4% to 100%. The atlas retains inactive responses: compact, spatially associated features do not guarantee broad coverage or faithful reconstruction.

### 6.3 Decoder features probe question-conditioned answer evidence

We probe five decoder configurations: Qwen2.5-VL-3B-Instruct, Qwen3.5-4B, LLaVA-NeXT Llama3-8B, LLaVA-v1.6-Mistral-7B and PaliGemma2-3B mix448. Each uses a published SAE with its native encoding equation and activation boundary (BeEnough 2026; Chanin and Lin 2026; LMMs-Lab 2024; Rajaram et al. 2025; Batra et al. 2026). All weights remain frozen. Qwen3.5's and PaliGemma's text-position SAEs are transferred to image-token states.

Three COCO photographs are paired with eight object-focused questions: two from VQAv2 and six constructed from visible attributes (Goyal et al. 2017). Figure 18 shows one question per photograph. Each heatmap sums local gradient-times-feature contributions to a fixed reference-versus-foil contrast of mean answer-token log probabilities. Up to eight feature IDs are selected by their positive contributions on that example, without using the object box. Maps retain positive net contributions and are normalized independently.

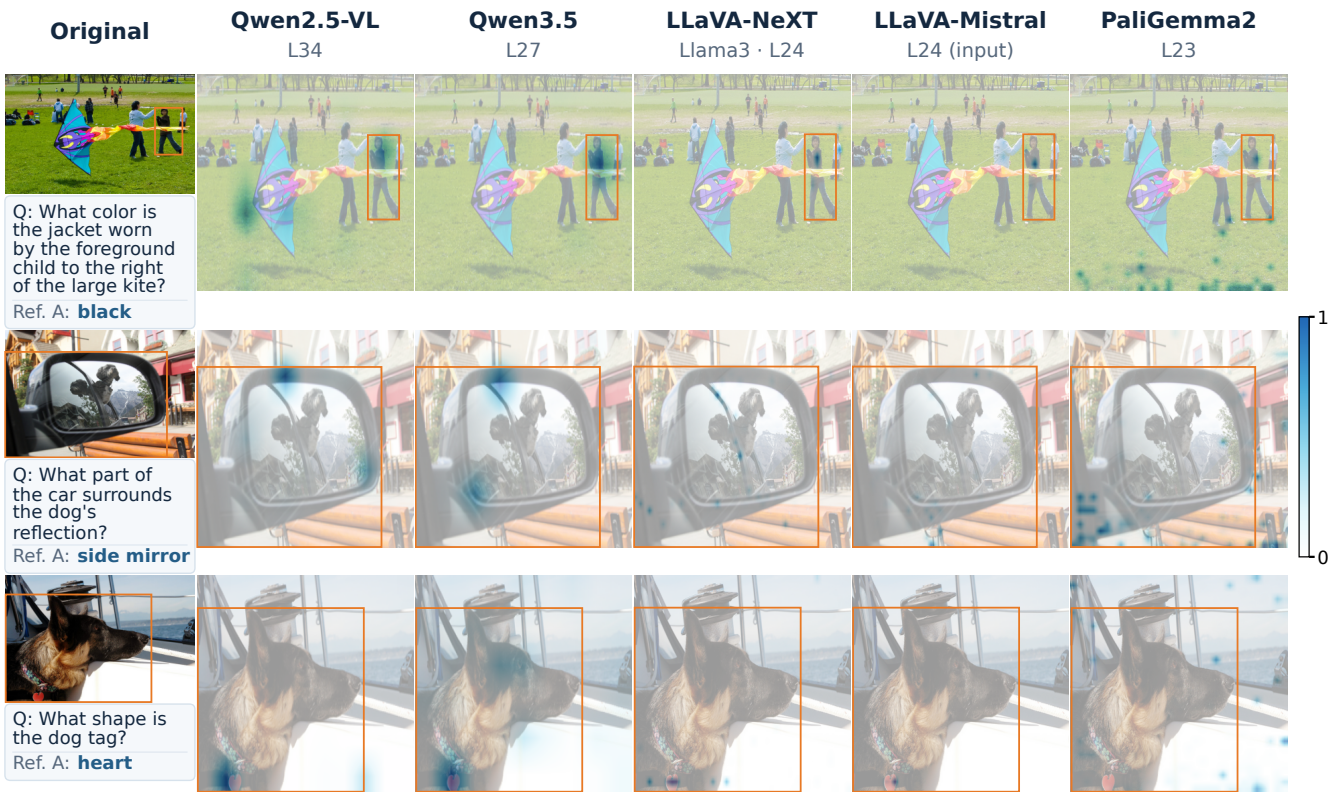



**Figure 18.** Question-conditioned decoder SAE attribution across five model configurations. Inputs and reference answers appear at left; heatmaps show positive net contributions from selected SAE feature groups. Orange boxes mark target objects; maps are normalized independently.

The same image supports different spatial patterns across decoder configurations and answer objectives. Qwen2.5's jacket map has a strong peak at the kite outside the target child's box, while its dog-tag map peaks near the collar and tag. These maps locate answer-sensitive SAE directions rather than segmenting the queried object. In Qwen and LLaVA, causal image-prefix states cannot read the later question; question conditioning enters through the downstream answer objective.

These case diagnostics reveal objective-dependent SAE directions, not a method ranking. Attribution establishes neither accurate generation nor faithful reconstruction: both LLaVA variants and PaliGemma answer "brown" to the jacket question, while the fixed contrast scores "black" against "white." PaliGemma's transferred dictionary also reconstructs image states poorly (FVU greater than one). Scoring and attribution definitions, feature-deletion results and reconstruction/pixel controls appear in Appendix A.7.

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# A Experimental details

## A.1 Checkpoints and input contracts

The shared VLM runs use Torch 2.6 and Transformers 5.3, BF16 weights, eager attention, and full-sequence uncached forwards. Pixtral uses a 256-pixel longest edge and Transformers' native Mistral regex correction, applied after tokenizer loading to avoid a duplicate-keyword bug in version 5.3. Pinned files are unchanged. Checkpoint identities are listed in Table 5; full revisions bind the model artifacts used by every archived run. Weights and bulk datasets are stored outside the repository under `/bigdata/gexin/vlm_probing`.

**Table 5.** Pinned pretrained checkpoints for the shared VLM experiments.

| Checkpoint | Revision |
|---|---|
| `HuggingFaceTB/SmolVLM-500M-Instruct` | `a7da5b986cb59b408707209984f360a5f4ad7e47` |
| `OpenGVLab/InternVL3-1B-hf` | `014c0583a0d4bedf29fbe2dbff4f865eb998e171` |
| `Qwen/Qwen3-VL-2B-Instruct` | `89644892e4d85e24eaac8bacfd4f463576704203` |
| `mistral-community/pixtral-12b` | `c2756cbbb9422eba9f6c5c439a214b0392dfc998` |
| `llava-hf/llava-1.5-7b-hf` | `b234b804b114d9e37bb655e11cbbb5f5e971b7a9` |
| `Qwen/Qwen3.5-4B` (appendix) | `851bf6e806efd8d0a36b00ddf55e13ccb7b8cd0a` |

GQA photographs come from the official image archive; questions and boxes come from the cross-modal information-flow authors' choose-attribute subset. The image IDs are 2341576 and 2342134, with boxes $(264, 33, 57, 89)$ and $(76, 147, 124, 134)$ in $(x, y, w, h)$ coordinates. MMVP uses the first released image pair. Its HF split is named `train`, but the images are released evaluation cases, not lens-fitting examples. VSR uses rows 0 and 1 of the zero-shot test split. The first row's car is partly visible as a dashboard, so it is unsuitable for a clear external-car localization claim. VisOnlyQA uses row 0 of the real geometry-triangle and chart-extraction splits, retaining its official no-reasoning prompts.

Shared-matrix GQA/MMVP/VSR prompts prescribe sorted answer options and no explanations. SmolVLM uses its native leading-space continuation; other word answers use their recorded bare-word encoding. LLaVA's Yellow has two answer pieces. The shared score includes all pieces. Original images, padded images, literal chat prompts, answer IDs, expanded positions, input hashes, and native generations are archived in `docs/report/v10/results`; previous records remain in `docs/report/results`.

## A.2 Hybrid attention in Qwen3.5

The preceding experiment also evaluated Qwen3.5-4B (`Qwen/Qwen3.5-4B`) on the eight shared inputs; its pinned revision is included in Table 5. Its 32 decoder blocks contain 24 Gated DeltaNet blocks and eight full-attention blocks. Residual lenses, activation patching, attribution patching, and steering remain defined at all block outputs. Local attention methods can inspect the full-attention blocks when the required probability sites are exposed. Standard attention rollout, however, assumes an attention matrix at every propagation step. Skipping DeltaNet blocks would omit their mixing and does not reproduce the standard method. Extending rollout requires defining and validating propagation through those blocks. Likewise, IOI-style path patching requires a controlled output-freezing contract for every component in the stack, which the current DeltaNet adapter does not supply. These structural boundaries differ from a missing editable softmax tap in a conventional decoder. The main comparison uses Pixtral to retain full-stack methods across all columns; the Qwen3.5 residual results remain in the archived supplementary experiment in `docs/report/results/qwen35`.

## A.3 Lens fitting and readout scope

The two VLM Tuned Lenses use 1,024 COCO2014 training images and 256 separate validation images, with eight caption prediction positions per image. The frozen model supplies 8,192 training and 2,048 validation positions. At layers 7, 15, and 23, FP32 affine translators minimize teacher-distribution KL with Adam, learning rate $3 \times 10^{-4}$, batch size 64, a maximum of 12 epochs, patience 3, improvement threshold $10^{-4}$, and seed 20261007. Epoch zero is a valid model-selection candidate. Shared report cases are excluded from fitting; recorded SmolVLM overlap checks include image hashes and VSR COCO source IDs. LLaVA artifacts were fitted previously using SDPA and image-mean padding, then replayed under the report's eager/gray conditions.

Table 6 reports independent validation fidelity. It records calibration quality on held-out caption positions, rather than ground-truth answer accuracy or a comparison of different lens objectives.

**Table 6.** Tuned Lens validation KL in nats per selected position. Initial and selected translators are evaluated on the same held-out caption positions.

| Model | Layer | Initial KL | Selected KL | Epoch |
|---|---|---|---|---|
| SmolVLM | 7 | 14.3279 | 1.9050 | 11 |
| SmolVLM | 15 | 11.4524 | 1.4967 | 12 |
| SmolVLM | 23 | 6.0436 | 0.7889 | 11 |
| LLaVA | 7 | 7.8680 | 1.3609 | 12 |
| LLaVA | 15 | 5.9005 | 0.9137 | 12 |
| LLaVA | 23 | 4.5529 | 0.3782 | 11 |

The GPT-2 Tuned Lens is imported from the authors and mapped to post-block sites. Attention Lens is fitted on 4,096 BookCorpus contexts (Zhu et al. 2015), with 512 WikiText-2 validation contexts for cross-domain validation (Merity et al. 2017). BookCorpus uses the historical HF archive's `books_large_p2.txt` prefix, not BookCorpusOpen; each context has 96 tokens and supervises the final position. WikiText uses `Salesforce/wikitext`, `wikitext-2-raw-v1`; its validation and test splits supply 512 contexts each. Jacobian Lens calibrates all twelve GPT-2 residual layers on 256 WikiText-2 training contexts. The preceding experiment selected only layers 3, 7, and 10; its uncalibrated rows were a configuration choice, not a method limitation. The Jacobian readout estimates causal transport into the final residual basis; its scores are not calibrated next-token probabilities. The available artifacts support the text study, not a five-VLM calibrated Jacobian/Attention Lens comparison.

EmbedLens reports lexical neighbors of projected visual input embeddings, without category thresholds for the full paper's token classification. EmbedLens and Patchscope retain the earlier single-word-or-phrase GQA instruction. Images and questions are shared, but their source residuals are not pooled with the matrix's forced-option run. Patchscope uses source layers 7, 15, 23, and 31 at the answer-prediction position, replacing the target placeholder after layer 0 with an identity mapping. Its zero-vector control uses the same target site and scoring prompt. Both are readout studies, not completed pruning or cross-model transfer trials.

## A.4 Attention implementations and spatial display

Single-layer Grad-CAM and ATTATTR use the last available native attention layer; decoder propagation methods start at layer 0 and require a consecutive stack. Qwen3-VL propagation does not include the vision encoder, projector, or DeepStack injections as separate graph edges. Qwen3.5 has eight full-attention layers at indices 3, 7, 11, 15, 19, 23, 27, and 31, interleaved with 24 DeltaNet layers. Skipping the latter is not a full-stack rollout.

TAM and Beyond input integrals use 20 right-Riemann samples at $\alpha = 1/20, \ldots, 1$. SmolVLM interprets exactly zero processed image pixels as a padding-image sentinel; excluding that endpoint avoids removing its image token sequence. This quadrature adaptation is applied consistently across the VLM comparison. ATTATTR retains the official-code left-sample convention with 20 samples. TAM uses residual normalization by $1/2$.

The raw native patch grid is interpolated in the recorded padded-image frame and then cropped using the original padding offsets. Each spatial map is normalized by its own native-grid maximum absolute value, without renormalization after cropping. In Figure 6, heatmaps remain fully opaque and the original photograph is overlaid at 20% opacity. Color intensity follows a monotone power transform with exponent 0.4 to make weak spatial variation visible. The palette runs from white at zero to blue for positive values and orange for negative ATTATTR values. This display transform does not change the archived scores; measured zero maps are explicitly labeled. Shirt-mass diagnostics integrate absolute values in the cropped display coordinates and are not benchmark localization scores. Original-grid numerical arrays remain available in the archives.

The native vision control uses the original timm/JAX checkpoint `jx_vit_base_p16_224-80ecf9dd.pth`, losslessly converted to a native HF 12-layer, $14 \times 14$-patch classifier. The Google ViT identifier supplies configuration and class labels, not the measured weights. ViT CAM/ATTATTR explain layer 11 at CLS; ATTATTR uses 20 right samples, TAM/Beyond use 20 endpoint-inclusive samples, and Beyond-H/C start at layers 3/5. Its cat–dog and elephant–zebra images are author-released illustrative assets, not a benchmark subset. Deletion sets selected processed patches to zero, equivalent to training-mean RGB; ten seeded random orders define the displayed envelope. Original ViT paper-specific attribution settings remain in their source records.

## A.5 Intervention contracts and validation

Residual patching uses an RGB$(128, 128, 128)$ reference image with identical dimensions and text. Sampled exact layers are 0, 5, 10, 16, 21, 26, and 31 on SmolVLM/LLaVA; attribution estimates use all 32 layers. Visual slots and answer-prediction slots are separate intervention sets. Token sweeps select question subwords and the final four assistant-boundary subwords. The public attribution API currently replays a gradient forward per layer; within that layer one gradient covers every token. This report does not claim the optimized all-sites two-forward/one-backward cache. Four matched runs require 658 forwards and 132 activation backwards, with 18.17 GiB peak allocation.

The knockout study uses the LLaVA chat instruction "Answer the question using a single word or phrase." Its question is shown in Figure 10. Each center $c$ jointly masks all heads in $[\max(0, c-4), \min(31, c+4)]$. Image keys, question queries, and the final prediction position retain native expanded coordinates. Its metric is $P(Y)$ for token ID 612, not full-word Yellow probability. Shirt/other masks contain 20/556 keys. Blocking connections already forbidden by the causal mask is a null control. The second case has baseline $P(Black) = 0.9982$ and a maximum image-to-question decrease of 9.43%.

Path patching uses the paired MMVP images with the same question and text layout. Sender outputs at the prediction position are replaced from the reference input; every other attention-head output is fixed to its base cache. MLP and LayerNorm recompute, the final receiver residual is cached, and a fresh base run injects only that receiver. Complete sweeps cover every declared sender head in the five models; SmolVLM, InternVL, Qwen3-VL, Pixtral, and LLaVA expose 480, 336, 448, 1,280, and 1,024 paths, respectively. Full head-output freezing is unavailable for Qwen3.5's DeltaNet stack. Selected paths are configured by the user, not automatically assigned semantic component names.

The GPT-2 EAP graph includes input, head, and MLP output messages into editable pre-LayerNorm Q/K/V branches, MLP inputs, and final readout. EAP uses the clean gradient; EAP-IG uses five left input-embedding samples $0, 0.2, 0.4, 0.6, 0.8$, preserving position embeddings and endpoint message differences. The author IOI CSV row is 475; its reference replaces the final Andrew occurrence with Jeremy. This graph experiment prepends BOS and reports its own original/reference margins, 6.1643/3.2137. It should not be numerically pooled with the separately prepared text-lens readout. Every retained-edge curve point is an actual forward with the recorded post-pruning edge count.

The VSV comparison uses BF16 LLaVA-1.5-7B, SmolVLM-500M and Qwen3-VL-2B on the original COCO validation image 310196. Its common user request is "Please help me describe the image in detail." Each model uses its native chat template. Directions are the image-bearing minus image-free last-context block residuals. The no-image input retains exactly the positive input's text IDs after removing its complete visual span and formatting. This preserves LLaVA's boundary whitespace instead of retokenizing a deleted image marker. Qwen's no-image input has explicit zero-based three-axis positions, avoiding reuse of an earlier image's RoPE offset.

All decoder block outputs are configured. Joint intervention adds the fixed direction at the original context-end and generated positions, then restores the original vector norm. The retained scales are 0, 0.005, 0.01, 0.02, 0.05, 0.10 and 0.17. Every generation uses greedy uncached full-prefix replay, a 128-token limit, and no repetition penalty. Native uncached baselines use the same full-sequence logits readout; the default HF last-position readout can change a BF16 argmax tie through matrix rounding. Smol's native parity reference retains its unmodified projected image features across every prefix; HF 5.3 otherwise removes its pixel inputs after prefill. The library generation itself replays the image at every step. Complete outputs, IDs, direction magnitudes, input contracts and zero/identity controls are archived. Figure excerpts retain at most 40 original words per cell with exact source offsets; selected visual checks are recorded separately. COCO reference captions use both skiing and snowboarding for this image, so that activity-label difference is not marked as an error.

Figure 14 reuses this LLaVA baseline's first 80 generated prefixes in every ranking condition. Its strength curves fix the prefix "The image captures a snowboarder skillfully riding"; snowboard/couch use their first pieces, rather than complete-word probabilities. The earlier FP16, manually formatted VSV runs remain historical records. The current study does not include the released implementation's PCA, normalized MLP-output rule or SLA. Repetition at large raw-residual scales characterizes this intervention configuration, rather than complete VISTA.

Same-input replacement and post-hook baseline restoration are exact in the retained shared and new paired runs. Final-site readouts differ from native BF16 logits by at most 0.125 because of selected-position matrix rounding; FP32 graph parity is within $1.26 \times 10^{-4}$. These numerical checks verify execution, not the uniqueness of the proposed interpretation. Newly added attribution/activation measurements preserve the archived input and baseline scores; plotting changes leave all source measurements unchanged.

## A.6 SAE checkpoints, splits and evaluation

### A.6.1 Pinned dictionaries and inference functions

The Pythia backbone is `EleutherAI/pythia-160m-deduped`, revision 582159a2dfe3e712a8d47ae83dec95ae3bde8e7e. Its safetensors SHA256 is bbaae5c00917b163baa499fc8eb64859ee0c850c5fdecfc32f4d70dc07213575. Language dictionaries come from https://huggingface.co/adamkarvonen/saebench_pythia-160m-deduped_width-2pow12_date-0108, revision ede0e8b18119ee7d77b0f21d928383a00c73843d. For each of `Standard`, `TopK`, `GatedSAE`, `JumpRelu`, `BatchTopK` and `MatryoshkaBatchTopK`, we use `resid_post_layer_8/trainer_0`, `trainer_2` and `trainer_4`. The full family-directory suffix is `_pythia-160m-deduped__0108`. These are sparsity settings, not independent training-seed repetitions; the selected source configurations record seed 0. Table 8 gives the source settings and all measured results.

The CLIP ViT-L/14-336 backbone is `openai/clip-vit-large-patch14-336`, revision 55ac000197133e202aa4e8bafcd01bf5a7da6e49, with weight SHA256 f51cb7b04a2afb04828b1b8e0edc34bbf082080d098b9874f5756e7b7ba8c807. Visual dictionaries come from https://huggingface.co/mateuszpach/sae-for-vlm, revision 270d13c7cc53e2ab8e887f938b0bdd22560e9f1d. The CLS releases are `batch_top_k_20_x16` and `matroyshka_batch_top_k_20_x16`, each using `trainer_0/checkpoints/ae_100000.pt`. The patch release is `matroyshka_batch_top_k_20_x64/random_k_2`, using `trainer_0/checkpoints/ae_200000.pt`. Their SHA256 values are, respectively, 7a17406c5199443745d2aeb76dbadb5e0a9c6083fb3163df2b5a5e8983585bf2, 105d21a4db64c70064a54ffc12ce14869a668980e048c8a777753f48a0dc9ee5 and e251e47c376a531b822c3c146d94b2a1e6a098f6edc8180ac60df205478420cc. The visual source does not supply a complete original training configuration for these snapshots; the loading metadata specifies a minimal inference configuration from the pinned tensor schema and author hook definition, code revision 39dff5bd6dea67fc3ef350bc7b2312e5fcfc1493.

Conversion preserves feature order, decoder-column scales, biases and saved thresholds. Pythia TopK uses per-token selection; BatchTopK and Matryoshka use their saved inference thresholds. Matryoshka group sizes are $(128, 256, 512, 1024, 2176)$ for the Pythia dictionaries, $(1024, 2048, 4096, 9216)$ for CLS and $(4096, 8192, 16384, 36864)$ for the patch dictionary. They are read from the source rather than replaced by a default grouping.

The 18 Pythia and three CLIP ViT-L/14-336 dictionary conversions are checked on 64 real native activations per checkpoint against independent implementations of the saved author tensor equations, pinned at 52b0c54ba92630cfb2ae007f020ed447d4a5ba9f. The largest relative L2 error is $2.45 \times 10^{-7}$ for encoding and $2.29 \times 10^{-7}$ for reconstruction; the largest absolute error is $2.44 \times 10^{-4}$. These comparisons retain raw saved feature magnitudes, before optional loader-side decoder renormalization. They establish numerical agreement for the declared functions, not a semantic interpretation of their features.

Language and CLIP ViT-L/14-336 forwards use float32 weights, Torch 2.6, Transformers 5.3, eager attention and uncached full-sequence execution. Native hook captures equal the corresponding cached Hugging Face hidden states on 32 real inputs: block 8 is `hidden_states[9]` in Pythia, and block 22 is `hidden_states[23]` in CLIP. The activation caches, checkpoint files and dataset selections have recorded SHA256 values.

### A.6.2 Cross-backbone patch dictionaries

The first four spatial checkpoints use block-11 residual outputs: ReLU and TopK for CLIP ViT-B/32, and ReLU and TopK for DINO ViT-B/16. These four have input dimension 768 and dictionary width 49,152. The CLIP backbone is `laion/CLIP-ViT-B-32-DataComp.XL-s13B-b90K`, revision f0e2ffa09cbadab3db6a261ec1ec56407ce42912; its source `open_clip_model.safetensors` has SHA256 3c00043509d2e3f35ec62bd85a643a393883dcda624d8208d034cc390c708297. The DINO backbone is `facebook/dino-vitb16`, revision f205d5d8e640a89a2b8ef0369670dfc37cc07fc2, with `pytorch_model.bin` SHA256 a064e36c67289caaa5c949c0b3f7f31a0fcbcba5721f5fa12419933ec1f4fe6e. OpenCLIP visual tensors are mapped strictly into the equivalent native CLIP architecture, retaining the source GELU function and all visual parameters.

These four SAEs are author-published ViT-Prisma ImageNet1k releases (Joseph et al. 2025; Gadre et al. 2023; Caron et al. 2021). Table 7 places their native observations beside the other spatial checkpoints. CLIP ReLU, DINO ReLU and DINO TopK were trained on CLS plus patch tokens; CLIP TopK used patch tokens only. The ReLU L1 coefficients are $10^{-5}$ for CLIP and $1.288194820853984 \times 10^{-6}$ for DINO. Their TopK budgets are 64 and 1,024, respectively. All four are evaluated on spatial patches only; CLS tokens are excluded from their feature maps. Neither inference preprocessing nor the published sparsity budget is retuned on COCO.

**Table 7.** Published spatial checkpoints and native observation scopes. $d$ is activation width and $m$ dictionary width. All layer numbers are zero-based source block indices. "Raw" means no extra SAE input normalization; DINOv2's backbone final normalization is part of its native observation. SAE++ L1 is a level of a jointly trained hierarchy; RA-SAE retains its nonlinear encoder and relaxed archetypal dictionary.

| Backbone | Recipe | $d$ | $m$ | Native observation / input normalization |
|---|---|---|---|---|
| CLIP B/32 | ReLU | 768 | 49,152 | Block 11 residual; raw |
| CLIP B/32 | TopK | 768 | 49,152 | Block 11 residual; per-token sample normalization |
| CLIP B/32 | Gated | 768 | 49,152 | Block 9 MLP output; raw |
| DINO B/16 | ReLU | 768 | 49,152 | Block 11 residual; per-token sample normalization |
| DINO B/16 | TopK | 768 | 49,152 | Block 11 residual; per-token sample normalization |
| CLIP L/14-336 | Matryoshka | 1,024 | 65,536 | Block 22 residual; raw |
| CLIP B/16 | BatchTopK | 768 | 24,576 | Block 11 residual; raw |
| SigLIP B/16 | BatchTopK | 768 | 24,576 | Block 11 residual; raw |
| DINOv2 B/14 | BatchTopK | 768 | 24,576 | Final-normalized block 11 patches |
| Qwen2.5-VL-3B | TopK | 2,048 | 65,536 | Decoder block 34 visual tokens; raw |
| Qwen3-VL-4B | SAE++ L1 | 1,024 | 20,000 | Vision block 23 pre-merge patches; raw |
| DINOv2 B/14 + registers | RA-SAE | 768 | 32,000 | Final-normalized block 11 patches; nonlinear encoder |

For the three normalized ViT-Prisma dictionaries, the source function uses per-token mean $\mu_h$ and sample standard deviation $s_h$, with correction 1:

$$u = \frac{h - \mu_h}{s_h + 10^{-5}}, \qquad \hat{h} = s_h D(E(u)) + \mu_h.$$

The decoder rescales by $s_h$, without adding the encoder's epsilon. The CLIP ReLU checkpoint uses the unnormalized activation instead. These functions follow ViT-Prisma code revision 64daf1425f7e5311844213fd0fb6f4127a30cd16; feature order, biases and decoder scales are preserved.

Each source checkpoint is named `weights.pt`. The pinned CLIP ReLU release uses revision 2b1cb17ffb8b06e7f45a699dfc97d417f9879be4 and weight SHA256 c4a92171b0503059a9aafc43a0d0006f2522602c70f95f71648ccd9c6aeb2136. The CLIP TopK release uses revision 310b1d1679c60f792d721f9605c957a5cb4b764f and weight SHA256 daee1b639c478d630825e025a8bc0b77be341dafa9525b32746c66c8e3599119. The DINO ReLU release uses revision 3e3ea0cdcde4186db8069f4fe0ce219e8693155a and weight SHA256 ba04901a22b0a8b37255246fec6eb8cdb8e0592e844754527ab6d9e7f679f8fd. The DINO TopK release uses revision 3a6493789ca56c3fc8e85c5c437c9d704321c95c and weight SHA256 8cfb15d2106fdb2e7e0c408a798b34a201a1ca5d4d5bded2bf24130adfd691ad.

Backbone verification compares all 12 native block outputs on four real COCO images against independent source tensor equations, both before and after the source's layer-normalization and value-bias folding. Relative L2 differences are below $1.55 \times 10^{-6}$, with absolute differences below $1.17 \times 10^{-4}$. The native capture is separately checked against Hugging Face block-11 hidden states on 32 images. Each SAE conversion is then checked on 64 real native activations against its pinned source equations: feature relative L2 errors are below $6.93 \times 10^{-8}$, reconstruction relative L2 errors below $7.10 \times 10^{-7}$, and absolute reconstruction errors below $1.15 \times 10^{-5}$. These checks establish the declared backbone binding and inference function before evaluating spatial behavior.

The Gated CLIP-B/32 dictionary is the published Prisma block-9 MLP release, revision 2b72983a021743c1b10ef3eba2b0123e2bbb4f36. Its `n_images_2600058.pt` has SHA256 64f38e1b82776607af893690a6beb7efa9711814679432af0f6813c0f1b10a4e. The source uses 768-dimensional MLP outputs, 49,152 features, an L1 coefficient of 0.7, and no extra activation normalization. Its encoder is

$$\begin{aligned} a &= (h - b_d)W_e + b_g, \\ m &= \max((h - b_d)(W_e \odot \exp(r)) + b_m, 0)\,, \qquad z = \mathbf{1}[a > 0] \odot m. \end{aligned}$$

with $\hat{h} = zW_d + b_d$. Here $\exp(r)$ scales encoder columns; the gate bias and magnitude bias remain distinct. The source was trained on CLS plus patches; this evaluation excludes CLS and captures the actual `vision_model.encoder.layers.9.mlp` output before residual addition. The release-era Prisma OpenCLIP loader, revision dedf59894c55

fd843df1aa95388979b91a7a10e4, explicitly disables LayerNorm folding and centering of writing weights. The native backbone retains those source coordinates. Independent original OpenCLIP block calculations on four real COCO images agree with the native MLP observation to relative L2 error $2.18 \times 10^{-6}$. The 196 patch features agree exactly with the author's Gated equations; reconstruction relative L2 error is $7.76 \times 10^{-8}$. The public model-bound SAE API also reproduces those feature values exactly. These checks bind the additional dictionary to its MLP observation rather than substituting a residual activation.

The three ViSAEBench dictionaries use the published `batchtopk_32x_k64/model.safetensors` snapshots (Gohil et al. 2026). Their encoder is $a = \max(W_e h + b_e, 0)$, with inference $z_j = a_j \mathbf{1}[a_j \geq \tau]$. The decoder dictionary retains its published learned multiplier and has no output bias. The saved threshold is held fixed, so training target $k = 64$ does not force 64 active features on every COCO patch. The source thresholds are 0.54186058 for CLIP, 0.63923049 for DINOv2 and 0.66128719 for SigLIP. An inclusive float32 source threshold is represented by its immediately preceding float32 value when the native implementation uses a strict inequality. This preserves the source boundary rather than estimating a new threshold from COCO.

The corresponding backbones (Oquab et al. 2023; Zhai et al. 2023) are `openai/clip-vit-base-patch16`, revision 57c216476eefef5ab752ec549e440a49ae4ae5f3; `facebook/dinov2-base`, revision f9e44c814b77203eaa57a6bdbbd535f21ede1415; and `google/siglip-base-patch16-224`, revision 7fd15f0689c79d79e38b1c2e2e2370a7bf2761ed. CLIP and SigLIP expose the raw output of vision block 11; DINOv2 applies its final backbone LayerNorm to that block's patch residual. CLS is excluded for CLIP and DINOv2; the selected SigLIP encoder has no CLS token. The DINOv2 release has no register tokens. Source extraction follows ViSAEBench code revision deed5c8b485186c75e2fe45d1b0a37a1f2402a17.

The CLIP SAE is `visaebench/clip-vitb16-saes`, revision db1a83e91af8e0c8066491f2ccd24ebf5e20b174, with weight SHA256 7f23b7cf40ad46b073853a9db64ec8fbca49ea9aa64e509d368a3fa3063f13e4. The DINOv2 SAE is `visaebench/dinov2-vitb14-saes`, revision fe01c6294a6c7ae8725427c651820ecf0202317d, with weight SHA256 bf51f18cc39affd25c170b04a09ddd537880505cb7a62e06fdc9c44d136e6625. The SigLIP SAE is `visaebench/siglip-vitb16-saes`, revision 229ff1e539047bcf5a5ae079721dd3842a694b5b, with weight SHA256 817f62c0af22adfd49dda3c4babb03b1b05a5b0e5e7e8a4c4d0f2b234241b3f4. Checks on real native patches compare the converted encoder and decoder against the independent source equations, including saved multipliers and threshold boundaries; actual processor pixels verify the annotation transform.

The Qwen2.5 dictionary is `BeEnough/SAE-Res-Qwen2.5-VL-3B-Instruct-W64K-L34-K100` (BeEnough 2026), revision 1a138768b0b74ea6844399f43f7a7ad1a69c0bff, with `adapter_model.safetensors` SHA256 a6dd50d69cb91846bf3bb65f89fa72f62609bd3f84af3916268ed64c0818685b. Its backbone is `Qwen/Qwen2.5-VL-3B-Instruct`, revision 66285546d2b821cf421d4f5eb2576359d3770cd3. The native observation is the output of `model.language_model.layers.34`, restricted to image-token positions. For $q = W_e(h - b_d) + b_e$, the author retains the 100 largest values of $q$ without rectification and sets all other entries to zero:

$$z_j = q_j \, \mathbf{1}[j \in \mathrm{TopK}_{100}(q)], \qquad \hat{h} = W_d z + b_d.$$

This is value-ranked selection, not selection by absolute magnitude. Selected negative coefficients would remain negative in native inference. On the captured COCO calibration and validation visual positions, no negative retained coefficients are observed, including the selected category-feature maps. The source reports training on 498,000 LLaVA-NeXT-Data examples but does not specify its training token mask; the evaluation restricts observations to visual positions and does not infer a training mask from that choice. The backbone uses source bfloat16 precision; SAE inference and source conversion checks use float32 representations of the captured activations. The fixed prompt is "Describe the main objects in this image." Native chat formatting and image/token merging are retained.

The Qwen3 dictionary is the level-1 component of `YusongZhao666/csae-ckpt` (Zhao et al. 2026), revision a5a4e40f150bda1de46a22bf045e8b8d8fd92a1f, with `ae.pt` SHA256 03ffd54714141d06612a617f02cf5367258bf9381b17fc151eee73c76b6b39ba. It binds to `model.visual.blocks.23` of `Qwen/Qwen3-VL-4B-Instruct`, revision ebb281ec70b05090aa6165b016eac8ec08e71b17. The source SAE++ implementation is pinned at 0dd9056affeb72d06b6af865f2fff4de103a3832. Its level-1 encoder uses the learned threshold 9.30319691 after rectification, with training target $k_1 = 6$. The published level-1 weights and biases, already rescaled by the author at saving, are used directly:

$$a = \max(W_e(h - b_d) + b_e, 0), \quad z_j = a_j \mathbf{1}[a_j > \tau], \quad \hat{h} = W_d z + b_d.$$

Although the frozen level-1 inference is represented by the native thresholded SAE primitive, its weights were trained jointly within the SAE++ hierarchy. The experiment does not treat level 2 as a second independent patch method or reproduce the complete hierarchical training objective. All 315 original vision tensors are loaded strictly

from the pinned source shard; the language model is not instantiated. Source vision computation uses float16, with float32 SAE encoding of captured block-23 activations. The observation is before spatial merging; grouped native patches are inverted to raster order for image-grid analysis without changing feature values.

The RA-SAE release (Fel et al. 2025) is `matybohacek/RA-SAE-DINOv2-32k`, revision 1e10a216938e112302b31e3bb2f69818e59a12a9. Its `RA-SAE-DINOv2-32k.pth` has SHA256 3cf20c7e8a97e063273e5116a049f965cb74799e2abded0c5e77f2ec30e27faa. This ImageNet1k-trained dictionary binds to `facebook/dinov2-with-registers-base`, revision a1d738ccfa7ae170945f210395d99dde8adb1805, whose `model.safetensors` has SHA256 7a6f7b3b9fa4b8732e707476a03cd6cdce210048582f21aafb7991c17d98e362. It is a different backbone from the register-free ViSAEBench model.

For the final-normalized 768-dimensional observation, the saved encoder is

$$a = \max\big(\mathrm{LN}_{32,000}(Ah + b), 0\big), \qquad z_j = a_j \mathbf{1}[a_j \geq 0.829].$$

LayerNorm acts across the 32,000 feature pre-activations, with its saved affine parameters and epsilon $10^{-5}$. The fixed threshold is the author inference wrapper's value, not a COCO-fitted sparsity target. The nonlinear encoder is preserved rather than replaced by a single affine projection. The saved relaxed archetypal decoder is materialized using its inference projection:

$$U_j = \frac{\max(W_j, 0)}{\sum_i \max(W_{ji}, 0) + 10^{-8}}, \qquad R'_j = R_j \min\left(1, \frac{1}{\|R_j\|_2}\right),$$
$$D = e^s(UC + R'), \qquad \hat{h} = D^\top z.$$

Here $C$ is the saved set of activation anchors, $R$ is the relaxation with unit row-norm bound, and $s$ is the saved scalar multiplier. Feature order and dictionary scales are unchanged. Independent row-wise checks of the materialized dictionary differ by at most $8.94 \times 10^{-8}$; native and source encoder feature values agree exactly. This frozen nonlinear dictionary supplies the same public `BaseSAE.encode/decode` evaluation contract as the other recipes.

The backbone binding is independently checked against Meta's original `dinov2_vitb14_reg4_pretrain.pth`, SHA256 73182a088cf94833c94b1666d1c99e02fe87e2007bff57b564fb6206e25dba71, and source revision 7764ea0f912e53c92e82eb78a2a1631e92725fc8. All 224 mapped Hugging Face tensors match the original checkpoint exactly; real-image patch outputs agree to relative L2 error $2.83 \times 10^{-6}$. The native sequence contains 261 positions: CLS, four registers and 256 raster-ordered patches. The evaluation selects positions 5–260 after final LayerNorm, so neither CLS nor register activations enter the spatial maps or patch reconstruction metrics.

### A.6.3 Text selection, metrics and probe controls

Pile evaluation uses the explicit `val.jsonl.zst` and `test.jsonl.zst` files from `monology/pile-uncopyrighted`, revision 3be90335b66f24456a5d6659d9c8d208c0357119. Stripped-text SHA256 values exclude repeated selected documents across calibration and test. From the first 4,000 selected unique documents per file, the first sufficient records with at least 1,024 tokens supply 128 calibration contexts and 512 test contexts. Each context consists of the first 1,024 tokens of one document; documents are not concatenated. Tokenization uses no added special tokens. Historical overlap with backbone or SAE training documents is not established, so this is same-corpus held-out-file evaluation, rather than a verified unseen-training-data test.

The calibration mean is calculated over all 131,072 calibration positions. Each test reconstruction uses exactly the same 524,288 native positions. Squared errors and the fixed-reference denominator are summed globally in float64. Feature frequencies count $z_j \neq 0$, including signed values; a feature unused on this split is reported as inactive on this split. The 64-context fidelity subset is fixed to the first eight test batches of eight contexts. We score every shifted next-token target, yielding $64 \times 1023 = 65{,}472$ predictions. If $L_{\text{orig}}$, $L_{\text{rec}}$ and $L_{\text{zero}}$ are original, reconstructed and native-zero-residual NLL, loss recovered is

$$R_{\text{loss}} = \frac{L_{\text{zero}} - L_{\text{rec}}}{L_{\text{zero}} - L_{\text{orig}}}.$$

The original NLL is 2.3461 nats per prediction and the zero-residual NLL is 11.9161. The zero control sets the selected native residual vectors to zero; it does not merely zero SAE features while retaining a decoder bias.

AG News uses `fancyzhx/ag_news`, revision eb185aade064a813bc0b7f42de02595523103ca4. For each of four classes, the official training file supplies 1,000 training and 250 development documents; the official test file supplies 250 test

documents. Selection uses seed 17 and excludes equal stripped-text hashes across all three splits. Residual caches use right EOS padding to 128 tokens, with padded positions excluded from pooling and evaluation. On 32 checked documents, replacing masked padding values leaves valid activations unchanged. Singleton versus padded-batch float32 execution has maximum absolute residual difference 0.00217 and relative L2 difference $2.39 \times 10^{-5}$; these kernel-rounding differences are retained in the provenance.

Each concept probe averages nonpadding token features within a document. For a given one-versus-rest class, features are ranked by absolute point-biserial correlation using training labels only. Training mean and population standard deviation define scaling. Validation AUROC selects $C \in \{0.01, 0.1, 1, 10\}$ for a liblinear logistic classifier; smaller $C$ breaks ties. No development or test observation influences feature selection or scaling. The development split tunes regularization; this is not a cross-validation estimate. The final metrics use the 1,000 untouched test documents. Table 9 gives the corresponding controls: 768 raw neurons, 128 training-fitted principal components and a seed-17 Gaussian projection into 4,096 column-normalized directions. Each representation receives the same per-class feature budget.

**Table 8.** All 18 pretrained language dictionaries. Rows within each family use source trainer indices 0, 2 and 4. $\lambda$ is the published sparsity penalty; JumpReLU settings are training targets, while reported $\overline{L}_0$ is measured on Pile. FVU uses the independent calibration mean, and $\Delta$NLL is in nats per prediction. AUROC uses one or five selected features separately for each AG News one-versus-rest class.

| Method | Source setting | $\overline{L}_0$ | FVU | $\Delta$NLL | AUROC@1 | AUROC@5 |
|---|---|---|---|---|---|---|
| ReLU + L1 | $\lambda = 0.012$ | 468.80 | 0.0399 | 0.1145 | 0.9120 | 0.9329 |
| ReLU + L1 | $\lambda = 0.020$ | 227.06 | 0.0690 | 0.2493 | 0.9073 | 0.9362 |
| ReLU + L1 | $\lambda = 0.040$ | 77.71 | 0.1078 | 0.5796 | 0.8869 | 0.9270 |
| TopK | $k = 20$ | 20.00 | 0.1081 | 0.5751 | 0.8412 | 0.9247 |
| TopK | $k = 80$ | 80.00 | 0.0659 | 0.2080 | 0.9002 | 0.9318 |
| TopK | $k = 320$ | 320.00 | 0.0182 | 0.0341 | 0.9239 | 0.9370 |
| Gated | $\lambda = 0.012$ | 464.43 | 0.0063 | 0.0106 | 0.9050 | 0.9351 |
| Gated | $\lambda = 0.024$ | 215.58 | 0.0375 | 0.0804 | 0.9050 | 0.9351 |
| Gated | $\lambda = 0.060$ | 46.33 | 0.0843 | 0.3219 | 0.9038 | 0.9232 |
| JumpReLU | target $L_0 = 20$ | 20.25 | 0.1148 | 0.6236 | 0.8685 | 0.9216 |
| JumpReLU | target $L_0 = 80$ | 80.81 | 0.0657 | 0.1998 | 0.9088 | 0.9294 |
| JumpReLU | target $L_0 = 320$ | 321.67 | 0.0192 | 0.0331 | 0.9029 | 0.9337 |
| BatchTopK | $k = 20$ | 19.84 | 0.1074 | 0.5512 | 0.8671 | 0.9254 |
| BatchTopK | $k = 80$ | 79.37 | 0.0648 | 0.1945 | 0.9106 | 0.9289 |
| BatchTopK | $k = 320$ | 319.13 | 0.0218 | 0.0380 | 0.9208 | 0.9380 |
| Matryoshka | $k = 20$ | 19.88 | 0.1166 | 0.6178 | 0.9065 | 0.9230 |
| Matryoshka | $k = 80$ | 79.80 | 0.0736 | 0.2196 | 0.9060 | 0.9294 |
| Matryoshka | $k = 320$ | 320.62 | 0.0262 | 0.0455 | 0.8962 | 0.9349 |

**Table 9.** Matched concept-probe controls on the same AG News documents. The three AUROC columns use one, two or five selected features per class; F1 uses five. PCA and standardization are fitted on training documents only.

| Representation | AUROC@1 | AUROC@2 | AUROC@5 | Macro F1@5 |
|---|---|---|---|---|
| Raw neurons (768) | 0.8506 | 0.8929 | 0.9249 | 0.8038 |
| PCA (128) | 0.8511 | 0.9105 | 0.9481 | 0.8363 |
| Random projection (4,096) | 0.8773 | 0.9105 | 0.9286 | 0.7952 |

### A.6.4 Image geometry and feature selection

The COCO subset contains 500 training and 1,000 validation images selected by a seed-17 shuffle of sorted official image IDs, before any model outputs are observed. The annotations archive SHA256 is 113a836d90195ee1f884e704 da6304dfaaecff1f023f49b6ca93c4aaae470268. The four block-11 dictionaries and the block-9 Gated MLP dictionary

use the source ViT-Prisma transform: bicubic resize of the shorter edge to 224 pixels, with antialiasing, followed by a $224 \times 224$ center crop. Crop offsets use torchvision's rounded half-difference, including round-half-to-even behavior. CLIP uses its source RGB channel normalization; DINO uses ImageNet means and standard deviations. Their patch grids are $7 \times 7$ and $14 \times 14$, respectively. Reconstruction excludes CLS and uses all patches: 24,500 CLIP or 98,000 DINO calibration vectors define each observation's reference mean; 49,000 CLIP or 196,000 DINO validation vectors define the reconstruction metrics. The Gated MLP dictionary uses its own block-9 reference mean and feature RMS, with the same 24,500 calibration and 49,000 validation patch positions. Feature selection and localization also use every patch. Raw maps are bilinearly resized to the native $224 \times 224$ category-box grid for localization, without per-image contrast normalization.

For the additional CLIP ViT-L/14-336 cohort, the slow processor resizes the shorter edge to 336 pixels, then applies a $336 \times 336$ center crop with offset $\lfloor (s - 336)/2 \rfloor$ along a resized dimension of size $s$. Boxes undergo the identical resize and crop, are clipped to the input bounds, and are unioned by category. Crowd boxes and boxes lying entirely outside the crop are excluded. This geometry is verified against processor pixels, including odd-sized resized dimensions; no square padding is used.

For this cohort, the selected activation has 577 positions: CLS followed by a $24 \times 24$ patch grid. CLS reconstruction uses position 0. Patch reconstruction uses all 576 spatial positions: 288,000 calibration vectors define the fixed reference mean and 576,000 validation vectors define FVU and activity. The spatial evaluation also uses every patch and bilinearly resizes the measured map to the $336 \times 336$ annotation grid, without per-image min–max normalization. CLS and patch reconstruction remain separate observations. Table 10 gives the native position counts for every backbone.

The ViSAEBench CLIP-B/16 processor uses bicubic shortest-edge resize to 224 and a centered $224 \times 224$ crop with floored offsets, producing $14 \times 14$ spatial tokens. DINOv2 uses the author's bicubic shortest-edge resize to 256, then torchvision's rounded center crop to 224, producing $16 \times 16$ patches. Its RGB normalization is ImageNet's. SigLIP directly resizes to $224 \times 224$, without a center crop, with channel mean and standard deviation 0.5; its grid is $14 \times 14$. Every category box receives the same resize and crop as the actual pixels. RA-SAE uses the same bicubic resize-to-256, rounded center crop to 224 and ImageNet normalization, with the register backbone's $16 \times 16$ spatial grid. Source crop differences can change the visible region and eligible pair membership even though the raw COCO image order is shared.

Both Qwen routes use their native fast image processors with minimum and maximum pixel budgets set to 65,536. Dimensions retain the source aspect ratio subject to the processor's patch-multiple rounding; no center crop is added. Qwen2.5 uses 14-pixel patches and spatial merge factor 2. Its $h \times w$ processor patch grid becomes an $(h/2) \times (w/2)$ visual-token grid in decoder block 34. Those positions are identified by native image-token IDs after chat and image/text merging. Qwen3 uses 16-pixel patches. Block 23 is observed before merge factor 2, and its grouped patch order is inverted to the native $h \times w$ raster grid. A small indexed-grid fixture checks this inverse for scalar and feature-valued maps. Recovered RGB images are reconstructed from the actual normalized processor tensors, so plotted grids, pixels and scaled boxes share the same coordinate frame. Variable image shapes are grouped by native grid and pixel shape when calling the shared localization pipeline.

For each additional checkpoint, the reference mean and feature RMS use every captured calibration visual position; reconstruction evaluates every captured validation visual position. Position counts therefore depend on native geometry and are retained with the per-model results. These all-patch measurements use the same reconstruction definition as the CLIP-L/14-336 patch evaluation above, while retaining each source grid and observation width.

For every dictionary, category $c$ and feature $j$, calibration selection maximizes

$$A_{cj} = \frac{\overline{z^+_{cj}}^{\text{inside}} - \overline{z^+_{cj}}^{\text{outside}}}{\sqrt{\mathbb{E}_{\text{cal}}[z_j^2]}}.$$

Here $z_j^+ = \max(z_j, 0)$ defines positive evidence; the denominator retains the raw feature RMS, including signed values. Inside and outside means use category-positive calibration images, with area-weighted box coverage at patch resolution. The RMS uses all calibration patches. A category needs at least ten calibration images; a candidate feature must have positive evidence in at least $\max(3, 0.1n_c)$ category-positive images and have positive contrast. The 20 most frequent eligible categories from the CLIP-B/32 calibration selection define a fixed category list for the expanded cohort. Each new dictionary selects its own feature on calibration data; categories that do not meet the same visibility, positive-activity and contrast conditions are recorded as ineligible rather than filled using test data. Features are frozen before validation. The four block-11 dictionaries select the same category set and are scored on

**Table 10.** Native spatial observation counts on the fixed 500 calibration and 1,000 validation images. CLS is excluded. The Qwen rows retain their actual image-dependent grids and observe different stages.

| Backbone / scope | Native grid | Calibration positions | Validation positions |
|---|---|---|---|
| CLIP B/32 | $7 \times 7$ | 24,500 | 49,000 |
| DINO B/16 | $14 \times 14$ | 98,000 | 196,000 |
| CLIP L/14-336 | $24 \times 24$ | 288,000 | 576,000 |
| CLIP B/16 | $14 \times 14$ | 98,000 | 196,000 |
| SigLIP B/16 | $14 \times 14$ | 98,000 | 196,000 |
| DINOv2 B/14 | $16 \times 16$ | 128,000 | 256,000 |
| DINOv2 B/14 + registers | $16 \times 16$ | 128,000 | 256,000 |
| Qwen2.5-VL decoder L34 | Variable, post-merge | 36,921 | 73,965 |
| Qwen3-VL vision L23 | Variable, pre-merge | 109,468 | 219,592 |

the same 1,534 validation pairs from 774 source images. The figures use the fixed person, car and dog categories from this selection. Image–category pairs are pooled for the headline metrics, so common categories contribute more pairs. The same feature may associate with more than one category. Table 11 reports every selected category for the CLIP ViT-L/14-336 cohort.

Within-image permutation controls independently shuffle each native-grid feature map using seed 17, preserving feature values and activity while removing their measured spatial arrangement. Point@1 includes inactive positive-evidence maps as misses. Activation mass uses $M_j^+$, is undefined when that positive channel is zero, and is averaged only over positive-active pairs, with coverage reported separately. Raw signed feature activity used for $\overline{L}_0$ can therefore differ from positive spatial response. The target is a category-box union, not a pixel-level segmentation mask.

For the cross-backbone figures, every source photograph is processed independently by each backbone's native image transform. Ground-truth boxes undergo the identical resize and crop, and feature maps retain their native patch grid. The same category feature is reused across images within each dictionary. Let $r_j = \sqrt{\mathbb{E}_{\text{cal}}[z_j^2]}$ be its calibration RMS. All displayed maps in this cohort are nonnegative and use $z_j/r_j$ on the fixed linear range 0–6, saturating larger values. The raw Qwen2.5 encoder remains unrectified even though its retained COCO coefficients are nonnegative. No individual test panel is rescaled to its own maximum. The heatmap is opaque and uses white at zero, then cyan and blue as activation increases. Its native RGB image is superimposed at opacity 0.40, followed by aligned orange GT boxes. Feature IDs, raw peaks and calibration values remain in the rendering records rather than as per-image labels. PDF export keeps the two image layers separate so image opacity is applied once. Inactive maps remain inactive.

The method-comparison figure uses the same fixed person photograph, COCO 9378, for both backbone rows: CLIP-B/32 DataComp and DINO-B/16. Its columns show the uncropped original followed by ReLU and TopK maps. All four dictionaries read block-11 residuals. The dictionaries retain their source-specific activation normalization and sparsity settings, rather than imposing an equal activity budget. The compact atlas has four rows: person 9378, person 14439, car 263463 and dog 532575. Its first column shows uncropped, full-color originals and original category boxes. Six subsequent columns retain one fixed visual-encoder dictionary per backbone: Matryoshka for CLIP-L/14-336; ReLU for DINO-B/16; BatchTopK for SigLIP and DINOv2 without registers; RA-SAE for DINOv2 with four registers; and SAE++ level-1 for Qwen3-VL-4B vision layer 23. Qwen3 is observed before patch merging and language decoding. The two DINOv2 columns use different backbone weights and token layouts, with register tokens excluded from the RA-SAE raster. The other evaluated checkpoints, including the distinct Qwen2.5 decoder observation, remain in the quantitative tables. That inventory spans observation stages; neither it nor the encoder-only atlas isolates an SAE algorithm while holding backbone weights and activation hooks fixed. Native crops can show different extents of the same source photograph. All four cases and their category features are reused from the completed evaluation; compact rendering does not rerun inference or choose new features. RA-SAE's two person maps and its dog map retain their measured zero response without a per-image text label; the selected car feature remains active. Rendering records retain original-image, native-image and map hashes, feature IDs, RMS values, native grids, aligned boxes and saturated-value fractions.

Sampling intervals use 2,000 paired percentile-bootstrap resamples with seed 17. Language comparisons resample the 64 source documents. The additional CLIP ViT-L/14-336 localization intervals resample the 774 source images contributing its 1,534 image–category pairs, retaining every paired row of a sampled image; category pairs from one image are not treated as independent observations. The BatchTopK-minus-ReLU NLL increase has interval $[-0.4192, -0.3569]$ nats, and the near-20-activity Matryoshka-minus-TopK difference is 0.0427 nats with interval $[0.0216, 0.0635]$. Real-minus-shuffled point@1 differs by 44.0 percentage points with interval 41.5–46.4; active-only mass differs by 42.1 points with interval 40.5–43.6. These intervals condition on fixed dictionaries, feature selection, splits and the recorded shuffle realization. They do not estimate SAE training-seed variability.

**Table 11.** Complete held-out localization results for the CLIP ViT-L/14-336 Matryoshka patch dictionary and its train-selected categories. $N$ counts image–category pairs. Coverage, point@1 and active-only mass inside category-box unions are percentages; inactive maps count as point@1 misses. Feature IDs preserve the author's dictionary order.

| Category | Feature | $N$ | Coverage | Point@1 | Mass (active) |
|---|---|---|---|---|---|
| Backpack | 370 | 40 | 92.5 | 62.5 | 42.4 |
| Bench | 3276 | 45 | 68.9 | 60.0 | 77.8 |
| Bottle | 2167 | 79 | 64.6 | 41.8 | 56.8 |
| Bowl | 574 | 56 | 89.3 | 71.4 | 67.4 |
| Car | 2499 | 97 | 100.0 | 75.3 | 52.5 |
| Cell phone | 1195 | 36 | 72.2 | 50.0 | 59.2 |
| Chair | 1848 | 113 | 100.0 | 70.8 | 51.2 |
| Clock | 2105 | 37 | 86.5 | 64.9 | 56.5 |
| Cup | 553 | 71 | 71.8 | 52.1 | 60.8 |
| Dining table | 2557 | 92 | 65.2 | 42.4 | 61.6 |
| Dog | 119 | 45 | 97.8 | 88.9 | 82.2 |
| Handbag | 6720 | 47 | 91.5 | 57.4 | 48.9 |
| Knife | 266 | 35 | 88.6 | 34.3 | 45.5 |
| Motorcycle | 2456 | 25 | 96.0 | 88.0 | 77.3 |
| Person | 420 | 533 | 84.2 | 68.9 | 77.4 |
| Pizza | 115 | 26 | 96.2 | 96.2 | 92.6 |
| Sink | 1420 | 44 | 100.0 | 77.3 | 49.1 |
| Surfboard | 2507 | 31 | 96.8 | 54.8 | 49.5 |
| Tie | 4301 | 35 | 8.6 | 0.0 | 0.0 |
| Truck | 2606 | 47 | 70.2 | 40.4 | 51.5 |

### A.6.5 Additional CLIP ViT-L/14-336 CLS dictionaries

Two 16,384-feature CLS dictionaries, BatchTopK and Matryoshka, use the same CLIP-L/14-336 block-22 backbone and fixed COCO image selections as the patch study. Their source dictionaries were trained on ImageNet. These single-vector observations assess representation retention beyond the spatial patch maps.

**Table 12.** CLS dictionaries on 1,000 held-out COCO images. FVU uses a reference mean from 500 calibration images. Cosine is measured at the final projected image embedding after native CLS reconstruction.

| Dictionary | Scope | Width | $\overline{L}_0$ | FVU | Cosine |
|---|---|---|---|---|---|
| BatchTopK | CLS | 16,384 | 24.35 | 0.2816 | 0.9709 |
| Matryoshka | CLS | 16,384 | 23.76 | 0.2956 | 0.9696 |

CLS reconstruction retains final projected image-embedding cosine similarity at 0.9709 for BatchTopK and 0.9696 for Matryoshka. This measures representation retention rather than classification accuracy. The all-patch Matryoshka results appear in Table 4; its complete category-level localization results remain in Table 11. Across 1,534 held-out image–category pairs, patch-feature coverage is 83.1%, point@1 counting inactive maps as misses is 62.5%, and shuffled point@1 is 18.5%. Conditional on positive activity, mass inside the target boxes is 64.9%, versus 22.8% after

spatial shuffling. The dictionary contributes the CLIP-L/14-336 maps in Figure 17, using the same independently selected feature IDs as in this quantitative evaluation.

### A.6.6 Reproducibility

The experiment scripts in `experiments/sae_report_v1` use the public `vlmprobe.evaluation.sae` API. Given the pinned raw files and weights, the staged replay is:

```
python experiments/sae_report_v1/prepare_data.py --text --coco
python experiments/sae_report_v1/capture_language.py
python experiments/sae_report_v1/verify_language_cache.py
python experiments/sae_report_v1/evaluate_language.py
python experiments/sae_report_v1/capture_vision.py
python experiments/sae_report_v1/evaluate_vision.py
python experiments/sae_report_v1/summarize_results.py
python experiments/sae_report_v1/estimate_uncertainty.py --resamples 2000 --seed 17
```

The cross-backbone comparison is maintained in `experiments/sae_visual_comparison_v2`. Its commands preserve the same raw COCO selection and consume pinned weights:

```
python experiments/sae_visual_comparison_v2/download_sources.py
python experiments/sae_visual_comparison_v2/run_experiment.py --model clip
python experiments/sae_visual_comparison_v2/run_experiment.py --model dino
python experiments/sae_visual_comparison_v2/collect_results.py
python experiments/sae_visual_comparison_v2/render_figures.py
```

The checkpoint verifier additionally accepts real native activation tensors through `--activation-dir`; the backbone preflight and the real-input conversion checks are retained with the measured results. The rendering stage reads measured maps and predeclared cases only; it does not load a model, select a feature or reorder images by a test score.

The expanded spatial evaluation uses `experiments/sae_visual_comparison_v3`, retaining the same raw COCO selection and fixed categories. With the pinned source weights in place, the additional model capture, calibration and evaluation stages are:

```
python experiments/sae_visual_comparison_v3/run_new_models.py --model all --phase all
python experiments/sae_visual_comparison_v3/evaluate_matryoshka_allpatch.py --stage all --device cuda:0
python experiments/sae_visual_comparison_v3/run_new_models.py --phase collect
```

Source equations and input geometry are verified before full capture. The native artifacts preserve each source recipe, including signed value-ranked TopK and the frozen SAE++ level-1 threshold. The additional Gated dictionary uses its independently verified MLP hook:

```
python experiments/sae_visual_comparison_v4/run_gated.py --phase all --device cuda:0
```

The RA-SAE route preserves the nonlinear encoder and register-token scope:

```
python experiments/sae_visual_comparison_v4/ra_sae.py download
python experiments/sae_visual_comparison_v4/ra_sae.py convert --device cuda:0
python experiments/sae_visual_comparison_v4/ra_sae.py preflight --device cuda:0
python experiments/sae_visual_comparison_v4/run_rasae.py all --device cuda:0
```

The completed source-bound results are collected once. Compact rendering reads the existing cache without rerunning inference or selecting features. The matched-backbone figure retains the version-6 artifact; the visual-encoder-only atlas uses version-7 selection and rendering:

```
python experiments/sae_visual_comparison_v4/collect_results.py
SAE_V4_OUTPUT=/bigdata/gexin/vlm_probing/outputs/sae_visual_comparison_v4
python experiments/sae_visual_comparison_v6/render_figures.py \
  --input "$SAE_V4_OUTPUT/visual_cases.pt" \
  --layout experiments/sae_visual_comparison_v6/layout.json \
  --case-selection /bigdata/gexin/vlm_probing/outputs/sae_visual_comparison_v2/display_selection.json \
  --output-dir /bigdata/gexin/vlm_probing/outputs/sae_visual_comparison_v6/figures
python experiments/sae_visual_comparison_v7/render_figures.py \
  --input "$SAE_V4_OUTPUT/visual_cases.pt" \
```

```
  --layout experiments/sae_visual_comparison_v7/layout.json \
  --case-selection /bigdata/gexin/vlm_probing/outputs/sae_visual_comparison_v2/display_selection.json \
  --output-dir /bigdata/gexin/vlm_probing/outputs/sae_visual_comparison_v7/figures
python experiments/sae_visual_comparison_v7/validate_figures.py
```

Final rendering reads only measured maps and predeclared display cases, rejects missing selected maps, and retains inactive measured responses. The atlas validator verifies all six visual-encoder bindings against the frozen source configurations, excludes the Qwen2.5 decoder dictionary and checks every map, image, aligned box and original photograph against the recorded payload.

Data, weights, native artifacts and generated outputs are stored under the existing `/bigdata/gexin/vlm_probing` hierarchy. The saved records include source revisions, checksums, raw sample IDs, selected masks and features, configuration, per-example scores and aggregate CSV/JSON results. The evaluation guide and live notebooks provide complete download-to-plot examples with smaller teaching subsets. Those tutorial scores are separate from the report measurements above. All reported checkpoints are fixed; comparisons do not estimate SAE training-seed variability.

## A.7 Pretrained multimodal decoder SAE protocol

The decoder case study uses five published dictionaries with their original weights and inference equations. The LLaVA-Llama3 dictionary accompanies K. Zhang et al. (2025); its training pipeline encodes complete multimodal conversations and flattens all decoder positions. The Qwen2.5 author reports 498,000 LLaVA-NeXT training samples without specifying an exact token mask. The Mistral dictionary comes from Rajaram et al. (2025). Qwen3.5's checkpoint-specific configuration records LMSYS-Chat-1M text conversations, 700 million training tokens and exclusion of special tokens (Chanin and Lin 2026). PaliGemma's MMDiff release trains on text positions of VQAv2 validation image/question forwards (Batra et al. 2026). The latter two dictionaries are transferred to visual positions here. In particular, the PaliGemma cases are not an independent held-out VQAv2 evaluation. Its released training script uses an `answer en` task prefix; this study uses the native plain question plus the common instruction below.

**Table 13.** Native pretrained decoder observations. All widths and sparsity budgets come from the published weights/configurations, not new training.

| VLM | Boundary | Hidden | Features | Source activation |
|---|---|---|---|---|
| Qwen2.5-VL-3B | L34 output | 2,048 | 65,536 | Raw TopK, $k = 100$ |
| Qwen3.5-4B | L27 output | 2,560 | 65,536 | JumpReLU export |
| LLaVA-NeXT Llama3-8B | L24 output | 4,096 | 131,072 | ReLU + TopK, $k = 256$ |
| LLaVA-v1.6-Mistral-7B | L24 input | 4,096 | 32,768 | ReLU, scale 8 |
| PaliGemma2-3B mix448 | L23 output | 2,304 | 16,384 | JumpReLU |

All layer indices are zero-based. The Mistral observation is the input to block 24, implemented at block 23's residual output; the LLaVA-Llama3 dictionary instead observes block 24's output. These leave 8 and 7 complete decoder blocks downstream, respectively; Qwen2.5, Qwen3.5 and PaliGemma leave 1, 4 and 2. No common layer number or relative depth is imposed.

Each run binds the following pinned backbone and dictionary releases:

- Qwen2.5-VL: Qwen2.5-VL-3B-Instruct and BeEnough's L34 TopK dictionary.
- Qwen3.5: Qwen3.5-4B and Decode Research's L27 Matryoshka BatchTopK dictionary.
- LLaVA-Llama3: LLaVA-NeXT Llama3-8B and LMMs-Lab's L24 dictionary.
- LLaVA-Mistral: LLaVA-v1.6-Mistral-7B and Line-of-Sight weights released in vlm-saes, file `sae_final_layer24_8x_norm13.996978.pkl`.
- PaliGemma: PaliGemma2-3B mix448 and MMDiff's text-position JumpReLU dictionary, file `text-only_layer_23.pt`.

Download manifests verify file sizes and source LFS SHA256 values. Frozen contracts bind the local weight files, code snapshots, questions, images and answer-format policies; checkpoints are never silently substituted.

**Native encoding and scale.** Qwen2.5 retains signed coefficients selected by raw-value TopK; LLaVA-Llama3 centers by decoder bias, applies ReLU and retains its largest 256 values. For Qwen3.5, the saved inference export computes $a = (h - b_d)W_e + b_e$, $z = \max(a, 0)\mathbf{1}[a > \theta]$, then $D(z) = zW_d + b_d$. It uses learned per-feature thresholds rather than reapplying the training budget of 100. PaliGemma uses the same thresholded form with $a = hW_e + b_e$: no decoder-bias centering or activation normalization is introduced.

Mistral follows the original published reconstruction-evaluation recipe: $z = \mathrm{ReLU}(\mathrm{fc1}(8h))$ and $D(z) = \mathrm{fc2}(z)/8$, retaining every positive coordinate. The later SAVE code uses a different capture boundary and a hard TopK filter; those choices are not substituted for this source recipe. Alternative public scale/boundary conventions are retained as preflight diagnostics, without choosing among them using QA outcomes.

**Question and box binding.** The eight fixed questions and canonical/foil strings appear in Table 14. Original VQAv2 validation questions join COCO2017 images by integer image ID to their COCO2014 counterparts. The dog-tag shape question is 532575002; the dog-breed question is 532575001. The tag-shape question has ten of ten answers "heart." The breed question has eight "german shepherd" answers, one "shepherd" and one "doberman"; that disagreement is retained. Official question and annotation ZIP archives are downloaded with byte-count, SHA256 and ZIP CRC checks. Constructed questions are visually checked investigator references rather than crowd annotations. Every model input appends the same instruction to the unmodified question:

> Answer with a short phrase in lowercase only. Do not write a full sentence.

The image remains unmarked. No reference answer, foil or box coordinate is given to free generation. Greedy generation has a 32-new-token ceiling. The canonical and foil are separately appended as answer content to the same scoring prefix; neither sequence includes an EOS/EOT in the scored content.

The score uses a fixed answer surface established from all eight native raw generations per model. Both Qwen models and PaliGemma start directly with lowercase content. Both LLaVA variants start with an initial capital; only LLaVA-Llama3 emits one preceding newline (token 198), which is shared context and is not scored. Mistral has no extra structural prefix. A three-surface audit retains lowercase, capitalized and leading-space lowercase scores before one rule is frozen for each route. This is a format check, not a selection of the highest-scoring spelling or a separate rule per question. Scores refer to these declared continuations rather than an invariant semantic answer posterior.

PaliGemma's native processor constructs a bidirectional image/question prefix with token type 0. Appended answer tokens have type 1 and use its causal suffix mask. A native tiny-model check verifies that changing a later answer leaves prefix states and earlier answer predictions unchanged; incorrectly retaining type 0 leaks that answer into image states. Changing the question can legitimately alter PaliGemma's image-prefix features. Both candidates use the identical question and attention convention.

**Table 14.** Fixed decoder questions. C denotes an investigator-constructed question; V denotes original VQAv2. The selected COCO instance ID binds all feature and pixel controls for that question.

| Image | Question | Canonical / foil | Instance | Source |
|---|---|---|---|---|
| 14439 | What color is the jacket worn by the foreground child to the right of the large kite? | black / white | 1733380 | C |
| 14439 | What color are the trousers worn by the foreground child to the right of the large kite? | blue / red | 1733380 | C |
| 263463 | What part of the car surrounds the dog's reflection? | side mirror / windshield | 145584 | C |
| 263463 | What color is the frame of the car's side mirror in the foreground? | black / white | 145584 | C |
| 263463 | What animal is reflected in the car's side mirror? | dog / cat | 10416 | C |
| 532575 | What shape is the dog tag? | heart / circle | 4318 | V |
| 532575 | What kind of dog is this? | german shepherd / poodle | 4318 | V |
| 532575 | What color is the heart-shaped tag on the dog collar? | red / blue | 4318 | C |

**Answer score and sparse attribution.** For image $I$ and question $Q$, let $A$ and $F$ denote the fixed reference and foil answer-token sequences. For any candidate $B$, $|B|$ counts its answer-content tokens. For the common image/question scoring prefix $P(I, Q)$, the candidate answer score and reference-versus-foil contrast are

$$\ell(B) = \frac{1}{|B|} \sum_{t=1}^{|B|} \log p(b_t \mid P(I, Q), b_{<t}), \qquad S = \ell(A) - \ell(F).$$

Only answer-content tokens enter the score; role markers, padding and end-of-turn tokens are excluded. Each answer token is scored at its preceding predictor position. Token counts, tokenwise log probabilities and summed scores are retained alongside the mean-token contrast. Length normalization makes $S$ a contrast of mean log probabilities, rather than complete-answer log odds. Reference continuations used for scoring are not evidence that the model freely generated that answer.

Let $z^B_{pf} = E(h^B_p)_f$ be the frozen source encoding at visual patch position $p$ in the pass for candidate $B$. An identity-valued proxy allows differentiation along dictionary directions without changing the unmodified forward. The signed feature-direction contrast and displayed map are

$$C^B_{pf} = z^B_{pf} \frac{\partial \ell(B)}{\partial z^B_{pf}}, \qquad C_{pf} = C^A_{pf} - C^F_{pf}, \qquad M^+_{p,\mathcal{J}} = \max\left( \sum_{f \in \mathcal{J}} C_{pf}, 0 \right).$$

For each model and case, $\mathcal{J}$ contains up to eight feature IDs with the largest positive global contributions $\sum_p C_{pf}$. This same-example, objective-selected group is exploratory; object boxes never select it. Figure 18 displays this exact group, which is also used for the measured deletion below. Canonical and foil continuations share the same scoring prefix but are differentiated separately. Signed contributions, negative evidence and the full-dictionary aggregate remain archived. With an affine decoder, the full-dictionary aggregate reduces to attribution along the reconstruction vector excluding decoder bias; the selected-group map retains its sparse decomposition.

These are local gradient attributions along SAE directions, rather than raw feature activations or attention weights. In Qwen and LLaVA, causal image-prefix states cannot read the later question; the downstream answer objective provides question conditioning. PaliGemma's bidirectional prefix can already encode the question, with answer leakage prevented by the causal suffix mask described above. Contextualized grid positions mix visual and decoder information: a peak is a token location, not an isolated pixel receptive field or an object-segmentation mask.

**Geometry and numerical validation.** Both Qwen models use their native fast image processor with minimum and maximum pixel budgets both set to 65,536. The post-merge decoder grid is reconstructed from `image_grid_thw`; displayed pixels are recovered by inverting native patch packing and normalization, using their respective patch and merge sizes. PaliGemma uses its native $448 \times 448$ RGB processor and $32 \times 32$ patch grid. Both LLaVA variants use native AnyRes processors. The retained cases produce a $30 \times 48$ or $32 \times 48$ high-resolution spatial grid; the 576-token low-resolution base crop and 30 or 32 newline tokens are excluded only from the displayed/intervened spatial branch. Original pixel coordinates pass through the actual resize, padding and unpadding affine. Native source packing is independently checked on five aspect ratios using coordinate-valued features. For display, RGB images, maps and boxes share a final affine resize to uniform tiles; native forward inputs and measurements remain unchanged.

The software environment uses PyTorch 2.6.0 (CUDA 12.4), Transformers 5.3.0 and Accelerate 1.15.0. All computations use frozen FP32 models with eager attention. LLaVA-Llama3 streams the weights of decoder blocks 0–23 from CPU storage to the GPU for execution; the observation block and downstream blocks remain on the GPU. This reduces resident weight memory without changing computation precision or the SAE equation. The three added routes run fully resident on separate GPUs. Native hook-output device checks verify GPU execution. The derivative is computed along the affine dictionary directions at active source coordinates, keeping inactive coordinates fixed. Canonical and foil passes retain their own native sparse coordinates. Their signed contributions are computed separately and subtracted after aggregation, preserving occasional TopK boundary changes from FP32 sequence-shape rounding. A separate equal-length masked-padding check yields identical visual-prefix states; the numerical guard records both maximum absolute difference and relative L2 difference. Its fixed acceptance bounds are $10^{-2}$ and $10^{-5}$, respectively; exceeding either rejects the case rather than replacing its source values. An identity-valued gradient proxy leaves forward hidden states bitwise unchanged. Source encoder/decoder equations, a sparse decoder's dense-equivalent forward and VJP, candidate token shifts and the first question's central finite difference are checked

before accepting results. No weights, thresholds or normalization statistics are updated. Canonical/foil prefix consistency, exact no-op candidate scores and the first question's feature-scaling finite difference with $\epsilon = 0.01$ are checked for every route. All forty accepted cases and the five directional checks pass. Full per-case reconstruction and perturbation controls remain separate from these numerical execution checks.

Table 15 evaluates reconstruction at the actual formal visual positions. Its FVU denominator centers hidden states across positions within that example, whereas the earlier aggregate SAE evaluation uses a training-only reference mean. Neither statistic measures the size of a dictionary-direction intervention.

**Table 15.** Dictionary fidelity at the image positions used in the illustrated decoder analyses. Values use the same formal image, question and answer-format instruction as the interventions. FVU is position-centered within the canonical pass; relative error is $\|D(z) - h\|_2 / \|h\|_2$, and mean $L_0$ counts active coordinates per image position.

| Model | Question | FVU | Relative error | Mean $L_0$ |
|---|---|---|---|---|
| Qwen2.5-VL-3B | Child jacket | 0.225 | 0.253 | 100.0 |
| Qwen2.5-VL-3B | Mirror part | 0.228 | 0.256 | 100.0 |
| Qwen2.5-VL-3B | Tag shape† | 0.221 | 0.254 | 100.0 |
| Qwen3.5-4B | Child jacket | 0.455 | 0.549 | 124.7 |
| Qwen3.5-4B | Mirror part | 0.489 | 0.563 | 118.0 |
| Qwen3.5-4B | Tag shape† | 0.407 | 0.534 | 124.9 |
| LLaVA-NeXT Llama3-8B | Child jacket | 0.106 | 0.273 | 256.0 |
| LLaVA-NeXT Llama3-8B | Mirror part | 0.105 | 0.275 | 256.0 |
| LLaVA-NeXT Llama3-8B | Tag shape† | 0.111 | 0.281 | 256.0 |
| LLaVA-NeXT Mistral-7B | Child jacket | 0.357 | 0.496 | 36.9 |
| LLaVA-NeXT Mistral-7B | Mirror part | 0.373 | 0.516 | 39.3 |
| LLaVA-NeXT Mistral-7B | Tag shape† | 0.385 | 0.518 | 36.0 |
| PaliGemma2-3B Mix-448 | Child jacket | 2.341 | 1.336 | 176.0 |
| PaliGemma2-3B Mix-448 | Mirror part | 2.364 | 1.363 | 181.4 |
| PaliGemma2-3B Mix-448 | Tag shape† | 3.040 | 1.488 | 183.1 |

**Feature and pixel intervention protocol.** For an edited sparse vector $z'$, the residual-preserving intervention is

$$h' = h + D(z') - D(z).$$

The native decoder $D$ returns raw activation units. This update retains the original reconstruction error. We remove group $\mathcal{J}$ at every valid spatial image-token position. The targeted effect $\Delta_8 = S_{\text{base}} - S_{\text{ablated}}$ measures the resulting change in the fixed answer contrast. At these late hooks, visual information already transferred into text positions remains intact; deletion tests the remaining downstream routes from image-token states.

Three seeded random controls sample active global feature IDs outside the targeted set when enough exist, using seed $17+$question index. Their mean score drop is $\overline{\Delta}_{\text{rnd}}$. The groups match the number of feature IDs, not their edited-coordinate counts or hidden-state perturbation norms. No-op and full-reconstruction controls are retained, and all coordinate counts and perturbation norms are recorded. Table 16 reports $S$, $\Delta_8$ and $\overline{\Delta}_{\text{rnd}}$ for all eight questions and five configurations.

Pixel controls fill the target COCO box with the rounded mean RGB of the original image and compare three same-area translations. The translated masks retain the integer-pixel size of the clipped target box; disjoint placements are preferred when feasible. The large car and dog boxes cover approximately 70.8% and 65.1% of their images, so equal-area translations necessarily overlap the target box. These are not background-only controls and cannot establish object-exclusive causal dependence. The dog tag has no separate COCO annotation and uses the whole-dog box.

**Observed local sensitivity.** For the jacket question, deleting the selected group changes the contrast from 0.649 to $-0.252$ in LLaVA-Mistral and from 0.083 to $-0.703$ in PaliGemma. Both nevertheless freely generate "brown" (with native capitalization), rather than the scored reference "black." Conversely, Qwen2.5 generates "black" for the trousers question despite a positive blue-versus-red contrast. These outcomes separate sensitivity of a fixed answer pair from unrestricted generation (Table 17).

Local gradient attribution and finite deletion can differ in magnitude: Qwen2.5's tag-shape group has signed first-order contribution 0.261, compared with a measured score drop of 0.572.

The transferred dictionaries also differ in visual reconstruction. Qwen3.5 has within-image FVU 0.407–0.489 on the three pictured inputs, whereas PaliGemma's FVU is 1.814–3.040 across the eight questions. The latter exceeds the within-image variance baseline, despite passing the native source-equation checks (Table 15). Its nonzero attribution and deletion effects therefore do not establish a faithful sparse decomposition of image states.

Random-ID controls match feature count but have much smaller perturbation budgets. For LLaVA-Llama3's jacket case, selected deletion has $\|\delta h\|_2 = 160.62$, versus a random-set mean of 5.82; their small score drops therefore do not establish semantic specificity. Reconstruction and pixel controls appear in Table 18; coordinate counts and perturbation norms appear in Table 19.

**Reproduction and saved evidence.** The original two routes reside in `experiments/sae_decoder_qa_v1/`; the three added routes are in `experiments/sae_decoder_qa_v2/`. The latter reuse the frozen v1 scoring and intervention engine through an explicit answer-format adapter. Native route modules load the pinned checkpoints. Completed questions resume only after the full source/data/options contract is checked. For example:

```
python experiments/sae_decoder_qa_v1/prepare_cases.py    --coco-dir /path/to/coco/val2017    --output outputs/decoder_
python experiments/sae_decoder_qa_v1/run_decoder_qa.py    --route qwen --cases outputs/decoder_qa/cases.json    --devi
python experiments/sae_decoder_qa_v1/run_decoder_qa.py    --route llava --cases outputs/decoder_qa/cases.json    --dev
PYTHONPATH=src python experiments/sae_decoder_qa_v2/score_format_probe.py    --route paligemma --cases outputs/decoder_
PYTHONPATH=src python experiments/sae_decoder_qa_v2/run_decoder_qa.py    --route paligemma --cases outputs/decoder_qa/
```

Each question saves the complete native prompt, tokenwise scores, generated answer, per-coordinate signed attribution, predictor-position features, reconstruction and feature-intervention effects, mask geometry and source hashes. Weak or negative effects are retained. Spatial plots normalize each positive net map separately and therefore compare arrangement rather than absolute evidence magnitude. The complete signed values remain in the saved tensors.

**Table 16.** Decoder feature deletions across eight fixed questions and five model configurations. † marks original VQAv2 questions; other questions are constructed.

| Model | Question | $S$ | $\Delta_8$ | $\overline{\Delta}_{\text{rnd}}$ |
|---|---|---|---|---|
| Qwen2.5-VL-3B | Child jacket | 4.972 | 0.008 | -0.0001 |
| Qwen2.5-VL-3B | Child trousers | 4.314 | 0.787 | 0.0006 |
| Qwen2.5-VL-3B | Mirror part | 1.982 | 0.027 | -0.0002 |
| Qwen2.5-VL-3B | Mirror frame | 5.274 | 0.713 | -0.0001 |
| Qwen2.5-VL-3B | Reflected animal | 5.725 | 0.229 | 0.0005 |
| Qwen2.5-VL-3B | Tag shape† | 2.452 | 0.572 | 0.0004 |
| Qwen2.5-VL-3B | Dog breed† | 5.820 | 0.107 | 0.0001 |
| Qwen2.5-VL-3B | Tag color | 5.177 | 0.024 | -0.0001 |
| Qwen3.5-4B | Child jacket | 5.601 | 0.217 | -0.0020 |
| Qwen3.5-4B | Child trousers | 4.981 | 0.126 | -0.0018 |
| Qwen3.5-4B | Mirror part | 0.621 | 0.209 | -0.0009 |
| Qwen3.5-4B | Mirror frame | 3.951 | 0.067 | 0.0008 |
| Qwen3.5-4B | Reflected animal | 3.093 | 0.201 | -0.0017 |
| Qwen3.5-4B | Tag shape† | 4.294 | 0.084 | -0.0018 |
| Qwen3.5-4B | Dog breed† | 5.037 | 0.215 | 0.0033 |
| Qwen3.5-4B | Tag color | 7.379 | 0.122 | -0.0022 |
| LLaVA-NeXT Llama3-8B | Child jacket | 0.334 | 0.551 | 0.0006 |
| LLaVA-NeXT Llama3-8B | Child trousers | 7.405 | 4.868 | 0.0003 |
| LLaVA-NeXT Llama3-8B | Mirror part | 2.132 | 0.230 | 0.0002 |
| LLaVA-NeXT Llama3-8B | Mirror frame | 5.843 | 0.278 | -0.0016 |
| LLaVA-NeXT Llama3-8B | Reflected animal | 8.339 | 0.419 | 0.0001 |
| LLaVA-NeXT Llama3-8B | Tag shape† | 7.207 | 0.243 | -0.0005 |
| LLaVA-NeXT Llama3-8B | Dog breed† | 5.811 | 0.114 | -0.0020 |
| LLaVA-NeXT Llama3-8B | Tag color | 9.743 | 0.604 | 0.0066 |
| LLaVA-NeXT Mistral-7B | Child jacket | 0.649 | 0.901 | -0.0001 |
| LLaVA-NeXT Mistral-7B | Child trousers | 7.208 | 0.362 | -0.0002 |
| LLaVA-NeXT Mistral-7B | Mirror part | 1.233 | 0.037 | 0.0000 |

Continued on next page

| Model | Question | $S$ | $\Delta_8$ | $\overline{\Delta}_{\text{rnd}}$ |
|---|---|---|---|---|
| LLaVA-NeXT Mistral-7B | Mirror frame | 5.902 | 0.104 | 0.0000 |
| LLaVA-NeXT Mistral-7B | Reflected animal | 7.742 | 0.372 | 0.0001 |
| LLaVA-NeXT Mistral-7B | Tag shape† | 6.936 | 0.164 | -0.0027 |
| LLaVA-NeXT Mistral-7B | Dog breed† | 4.908 | 0.135 | 0.0000 |
| LLaVA-NeXT Mistral-7B | Tag color | 7.776 | 0.534 | -0.0019 |
| PaliGemma2-3B Mix-448 | Child jacket | 0.083 | 0.786 | -0.0017 |
| PaliGemma2-3B Mix-448 | Child trousers | 6.306 | 1.352 | -0.0010 |
| PaliGemma2-3B Mix-448 | Mirror part | 1.979 | 0.609 | 0.0000 |
| PaliGemma2-3B Mix-448 | Mirror frame | 5.261 | 1.247 | 0.0013 |
| PaliGemma2-3B Mix-448 | Reflected animal | 8.481 | 0.957 | 0.0013 |
| PaliGemma2-3B Mix-448 | Tag shape† | 8.743 | 1.086 | 0.0036 |
| PaliGemma2-3B Mix-448 | Dog breed† | 4.390 | 0.359 | 0.0006 |
| PaliGemma2-3B Mix-448 | Tag color | 8.524 | 1.484 | 0.0011 |

**Table 17.** Greedy generated content from the original image and question. Display whitespace is trimmed; candidate scoring is separate.

| Question | Qwen2.5-VL | Qwen3.5 | LLaVA Llama3 | LLaVA Mistral | PaliGemma |
|---|---|---|---|---|---|
| Child jacket | black | black | Brown | Brown | brown |
| Child trousers | black | blue | Blue | Blue | blue |
| Mirror part | mirror | mirror | Mirror | Mirror | mirror |
| Mirror frame | black | black | Black | Black | black |
| Reflected animal | dog | dog | Dog | Dog | dog |
| Tag shape† | heart | heart | Heart | Heart | heart |
| Dog breed† | german shepherd | german shepherd | German shepherd | German shepherd | german shepherd |
| Tag color | red | red | Red | Red | red |

**Table 18.** Reconstruction and pixel controls. FVU is centered over canonical-pass image positions within each example. Score drops compare full reconstruction, target-box masking and the mean of three same-area translations. Max overlap is the largest translated-mask fraction overlapping the target.

| Model | Question | FVU | $\Delta_{\text{rec}}$ | $\Delta_{\text{box}}$ | $\overline{\Delta}_{\text{px}}$ | Max overlap |
|---|---|---|---|---|---|---|
| Qwen2.5-VL-3B | Child jacket | 0.225 | 0.001 | 2.766 | -0.200 | 0.000 |
| Qwen2.5-VL-3B | Child trousers | 0.225 | 0.006 | 0.789 | 0.468 | 0.000 |
| Qwen2.5-VL-3B | Mirror part | 0.228 | 0.010 | 2.169 | 2.234 | 0.926 |
| Qwen2.5-VL-3B | Mirror frame | 0.228 | -0.011 | 1.808 | 1.386 | 0.854 |
| Qwen2.5-VL-3B | Reflected animal | 0.228 | -0.021 | 4.627 | 0.229 | 0.000 |
| Qwen2.5-VL-3B | Tag shape† | 0.221 | -0.002 | 3.238 | 0.314 | 0.690 |
| Qwen2.5-VL-3B | Dog breed† | 0.221 | 0.010 | 5.174 | 5.016 | 0.942 |
| Qwen2.5-VL-3B | Tag color | 0.221 | 0.001 | 5.175 | 1.747 | 0.783 |
| Qwen3.5-4B | Child jacket | 0.455 | 0.013 | 6.490 | 0.464 | 0.000 |
| Qwen3.5-4B | Child trousers | 0.455 | -0.047 | -0.246 | 0.026 | 0.000 |
| Qwen3.5-4B | Mirror part | 0.489 | 0.102 | 1.793 | 1.821 | 0.926 |
| Qwen3.5-4B | Mirror frame | 0.489 | 0.096 | 1.529 | 0.731 | 0.854 |
| Qwen3.5-4B | Reflected animal | 0.489 | 0.036 | 0.889 | 0.074 | 0.000 |
| Qwen3.5-4B | Tag shape† | 0.407 | 0.105 | 6.382 | 0.371 | 0.690 |
| Qwen3.5-4B | Dog breed† | 0.407 | 0.114 | 4.693 | 4.617 | 0.942 |
| Qwen3.5-4B | Tag color | 0.407 | 0.133 | 6.852 | 1.278 | 0.783 |
| LLaVA-NeXT Llama3-8B | Child jacket | 0.106 | -0.077 | 2.385 | -0.054 | 0.000 |
| LLaVA-NeXT Llama3-8B | Child trousers | 0.106 | 0.159 | 1.592 | -0.217 | 0.000 |
| LLaVA-NeXT Llama3-8B | Mirror part | 0.105 | 0.006 | 2.200 | 2.068 | 0.926 |
| LLaVA-NeXT Llama3-8B | Mirror frame | 0.105 | -0.012 | 0.730 | 1.302 | 0.854 |
| LLaVA-NeXT Llama3-8B | Reflected animal | 0.105 | -0.050 | 0.150 | 0.012 | 0.000 |
| LLaVA-NeXT Llama3-8B | Tag shape† | 0.111 | 0.067 | 9.099 | 0.221 | 0.690 |
| LLaVA-NeXT Llama3-8B | Dog breed† | 0.111 | -0.001 | 5.029 | 5.424 | 0.942 |
| LLaVA-NeXT Llama3-8B | Tag color | 0.111 | 0.168 | 9.215 | 1.280 | 0.783 |
| LLaVA-NeXT Mistral-7B | Child jacket | 0.357 | -0.104 | 3.475 | -0.005 | 0.000 |
| LLaVA-NeXT Mistral-7B | Child trousers | 0.357 | 0.479 | -0.188 | -0.251 | 0.000 |
| LLaVA-NeXT Mistral-7B | Mirror part | 0.373 | 0.007 | 1.135 | 1.502 | 0.926 |

Continued on next page

| Model | Question | FVU | $\Delta_{\text{rec}}$ | $\Delta_{\text{box}}$ | $\overline{\Delta}_{\text{px}}$ | Max overlap |
|---|---|---|---|---|---|---|
| LLaVA-NeXT Mistral-7B | Mirror frame | 0.373 | 0.202 | 0.059 | 0.463 | 0.854 |
| LLaVA-NeXT Mistral-7B | Reflected animal | 0.373 | 0.595 | 1.266 | 0.339 | 0.000 |
| LLaVA-NeXT Mistral-7B | Tag shape† | 0.385 | 0.010 | 9.467 | 0.777 | 0.690 |
| LLaVA-NeXT Mistral-7B | Dog breed† | 0.385 | 0.015 | 3.922 | 3.861 | 0.942 |
| LLaVA-NeXT Mistral-7B | Tag color | 0.385 | 0.414 | 7.531 | 0.655 | 0.783 |
| PaliGemma2-3B Mix-448 | Child jacket | 2.341 | -0.032 | 3.445 | -0.057 | 0.000 |
| PaliGemma2-3B Mix-448 | Child trousers | 2.157 | -0.100 | 1.616 | -0.069 | 0.000 |
| PaliGemma2-3B Mix-448 | Mirror part | 2.364 | -0.406 | 1.189 | 1.293 | 0.926 |
| PaliGemma2-3B Mix-448 | Mirror frame | 1.814 | -0.185 | 3.815 | 2.202 | 0.854 |
| PaliGemma2-3B Mix-448 | Reflected animal | 2.538 | -0.361 | 1.808 | -0.014 | 0.000 |
| PaliGemma2-3B Mix-448 | Tag shape† | 3.040 | 0.127 | 7.655 | 0.503 | 0.690 |
| PaliGemma2-3B Mix-448 | Dog breed† | 2.118 | 0.144 | 3.980 | 3.795 | 0.942 |
| PaliGemma2-3B Mix-448 | Tag color | 2.523 | 0.370 | 7.654 | 1.055 | 0.783 |

**Table 19.** Actual deletion budgets. Each entry reports canonical/foil passes. $n$ counts edited nonzero coordinates across spatial positions; random columns average three groups. ID-count matching does not match coordinate count or perturbation energy.

| Model | Question | $n_8$ | $\bar{n}_{\text{rnd}}$ | $\|\delta h_8\|_2$ | $\overline{\|\delta h_{\text{rnd}}\|_2}$ |
|---|---|---|---|---|---|
| Qwen2.5-VL-3B | Child jacket | 207/207 | 14.3/14.3 | 652.70/652.70 | 46.65/46.65 |
| Qwen2.5-VL-3B | Child trousers | 295/295 | 16.7/16.7 | 1767.97/1767.97 | 54.76/54.76 |
| Qwen2.5-VL-3B | Mirror part | 296/296 | 16.0/16.0 | 901.80/901.80 | 41.48/41.48 |
| Qwen2.5-VL-3B | Mirror frame | 204/204 | 19.3/19.3 | 1825.23/1825.23 | 32.23/32.23 |
| Qwen2.5-VL-3B | Reflected animal | 119/119 | 16.7/16.7 | 1022.65/1022.65 | 32.66/32.66 |
| Qwen2.5-VL-3B | Tag shape† | 282/282 | 14.0/14.0 | 2148.47/2148.47 | 22.29/22.29 |
| Qwen2.5-VL-3B | Dog breed† | 74/74 | 13.7/13.7 | 220.78/220.78 | 25.55/25.55 |
| Qwen2.5-VL-3B | Tag color | 239/239 | 15.7/15.7 | 935.88/935.88 | 38.56/38.56 |
| Qwen3.5-4B | Child jacket | 170/170 | 26.0/26.0 | 41.15/41.15 | 7.75/7.75 |
| Qwen3.5-4B | Child trousers | 249/249 | 14.0/14.0 | 104.73/104.73 | 4.03/4.03 |
| Qwen3.5-4B | Mirror part | 101/101 | 18.3/18.3 | 36.60/36.60 | 7.60/7.60 |
| Qwen3.5-4B | Mirror frame | 139/139 | 14.0/14.0 | 42.56/42.56 | 7.20/7.20 |
| Qwen3.5-4B | Reflected animal | 187/187 | 18.3/18.3 | 112.95/112.95 | 7.29/7.29 |
| Qwen3.5-4B | Tag shape† | 203/203 | 21.7/21.7 | 51.81/51.81 | 8.05/8.05 |
| Qwen3.5-4B | Dog breed† | 56/56 | 23.7/23.7 | 20.37/20.37 | 10.51/10.51 |
| Qwen3.5-4B | Tag color | 130/130 | 20.0/20.0 | 80.71/80.71 | 6.78/6.78 |
| LLaVA-NeXT Llama3-8B | Child jacket | 5588/5588 | 152.7/152.7 | 160.62/160.62 | 5.82/5.82 |
| LLaVA-NeXT Llama3-8B | Child trousers | 9514/9514 | 90.3/90.3 | 268.38/268.38 | 3.76/3.76 |
| LLaVA-NeXT Llama3-8B | Mirror part | 9256/9256 | 71.0/71.0 | 239.33/239.33 | 3.42/3.42 |
| LLaVA-NeXT Llama3-8B | Mirror frame | 8577/8577 | 127.3/127.3 | 270.28/270.28 | 6.01/6.01 |
| LLaVA-NeXT Llama3-8B | Reflected animal | 5790/5790 | 112.0/112.0 | 248.22/248.22 | 4.08/4.08 |
| LLaVA-NeXT Llama3-8B | Tag shape† | 8087/8087 | 89.7/89.7 | 249.95/249.95 | 2.15/2.15 |
| LLaVA-NeXT Llama3-8B | Dog breed† | 5709/5709 | 112.0/112.0 | 230.22/230.22 | 5.28/5.28 |
| LLaVA-NeXT Llama3-8B | Tag color | 6860/6860 | 299.0/299.0 | 172.82/172.82 | 8.40/8.40 |
| LLaVA-NeXT Mistral-7B | Child jacket | 1165/1165 | 61.7/61.7 | 107.33/107.33 | 3.94/3.94 |
| LLaVA-NeXT Mistral-7B | Child trousers | 1811/1811 | 47.3/47.3 | 114.37/114.37 | 3.14/3.14 |
| LLaVA-NeXT Mistral-7B | Mirror part | 958/958 | 45.0/45.0 | 34.99/34.99 | 2.43/2.43 |
| LLaVA-NeXT Mistral-7B | Mirror frame | 513/513 | 50.7/50.7 | 19.31/19.31 | 3.93/3.93 |
| LLaVA-NeXT Mistral-7B | Reflected animal | 375/375 | 50.3/50.3 | 50.46/50.46 | 3.64/3.64 |
| LLaVA-NeXT Mistral-7B | Tag shape† | 923/923 | 54.3/54.3 | 33.65/33.65 | 3.13/3.13 |
| LLaVA-NeXT Mistral-7B | Dog breed† | 2059/2059 | 110.0/110.0 | 127.82/127.82 | 5.29/5.29 |
| LLaVA-NeXT Mistral-7B | Tag color | 433/433 | 55.3/55.3 | 21.84/21.84 | 3.46/3.46 |
| PaliGemma2-3B Mix-448 | Child jacket | 1731/1731 | 98.7/98.7 | 9448.96/9448.96 | 269.98/269.98 |
| PaliGemma2-3B Mix-448 | Child trousers | 1798/1798 | 100.7/100.7 | 9603.61/9603.61 | 275.84/275.84 |
| PaliGemma2-3B Mix-448 | Mirror part | 1073/1073 | 156.0/156.0 | 4625.68/4625.68 | 399.84/399.84 |
| PaliGemma2-3B Mix-448 | Mirror frame | 1607/1607 | 96.7/96.7 | 8144.84/8144.84 | 256.72/256.72 |
| PaliGemma2-3B Mix-448 | Reflected animal | 1430/1430 | 180.3/180.3 | 6554.28/6554.28 | 363.05/363.05 |
| PaliGemma2-3B Mix-448 | Tag shape† | 1445/1445 | 87.3/87.3 | 2289.34/2289.34 | 255.29/255.29 |
| PaliGemma2-3B Mix-448 | Dog breed† | 1137/1137 | 127.7/127.7 | 4885.53/4885.53 | 331.54/331.54 |
| PaliGemma2-3B Mix-448 | Tag color | 2049/2049 | 173.0/173.0 | 9603.87/9603.87 | 372.35/372.35 |

## A.8 Replaying figures and measurements

The archived lens, attention and causal report notebook replays bundled results and figures without weights or a GPU. The live SAE evaluation notebooks instead acquire raw inputs, load published checkpoints and compute new measurements; their smaller teaching subsets are separate from the report results. The following archive commands regenerate the earlier comparisons; dependencies include Pandoc, XeLaTeX, and Biber:

```
python examples/render_crossmodel_report.py \
  --source-dir docs/report/v10/results --output-dir docs/report/v11/lens/v2 \
  --case-ids gqa_07302654 --figure-types logit-lens
python examples/render_report_lenses_v09.py
python examples/render_report_attention_v09.py
python examples/render_report_attention_v09.py \
  --output docs/report/v11/attention/v2 --groups local --overlay-style contrast
python examples/render_report_causal_comparison.py \
  --output-dir docs/report/v11/causal/v2
python examples/render_report_vsv.py
python examples/render_causal_report.py
python examples/run_report_path.py --render-only \
  --source docs/report/v10/results/path --report-dir docs/report/v10/causal
python examples/build_technical_report.py
```

The measurement runners handle the shared matrix, attention comparison, paired residual replacements, and path sweeps. Their filenames and calls are linked in the report notebook; dedicated method notebooks retain data acquisition and inference calls. Calibrated artifacts bind model, tokenizer, activation site, and data provenance. Portable JSON and figure manifests record source hashes and exported-result equality. These artifacts support replay and new-input analysis while keeping bulk checkpoints outside the code repository.

# B Additional comparisons

## B.1 A second vision-classifier comparison

Figure 19 repeats the native ViT comparison on the author's elephant–zebra image with the same deletion protocol as the cat–dog control in Figure 7.

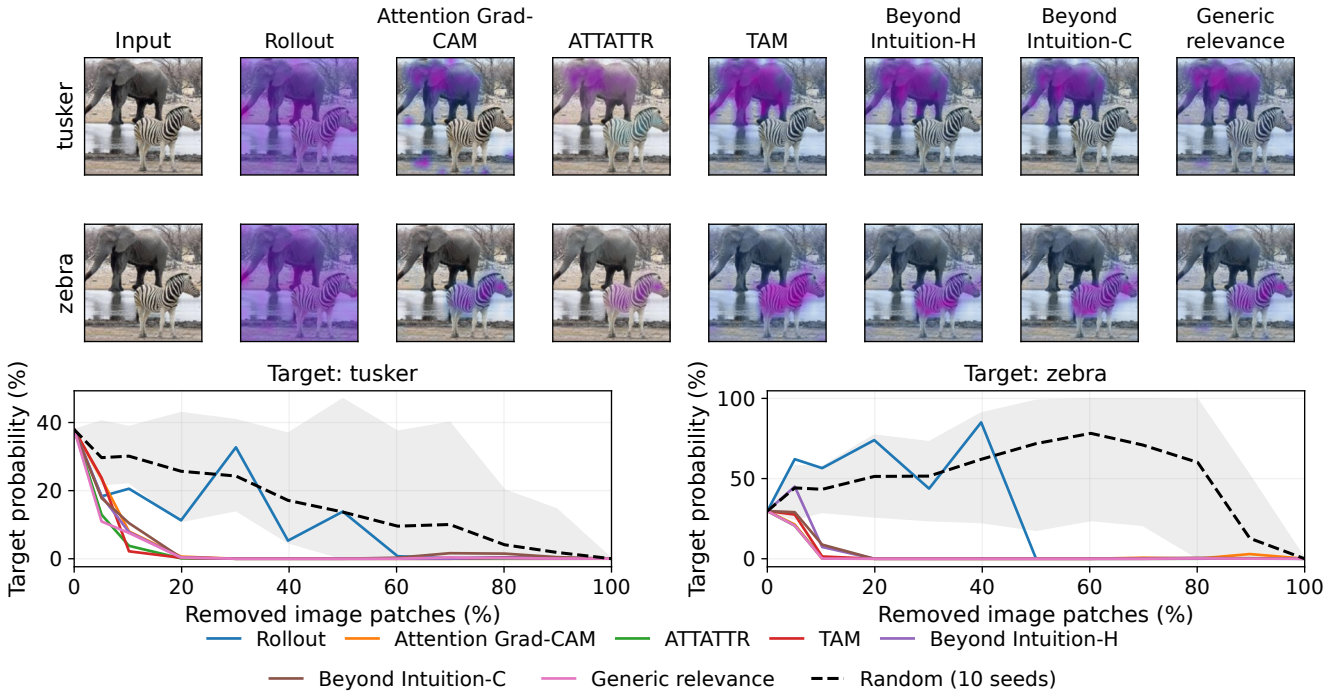



**Figure 19.** A second native ViT control separates elephant and zebra attribution. Identical image preprocessing and deletion budgets allow map differences to be checked against the effect of removing selected patches.

## B.2 Residual replacement on the black-shirt input

Figures 20 and 21 retain the second GQA case under the same reference input, sites, and metrics as the yellow-shirt study. The smaller LLaVA effect and stronger SmolVLM question-word sensitivity illustrate why the main comparison is interpreted per case rather than as a universal pathway ranking.

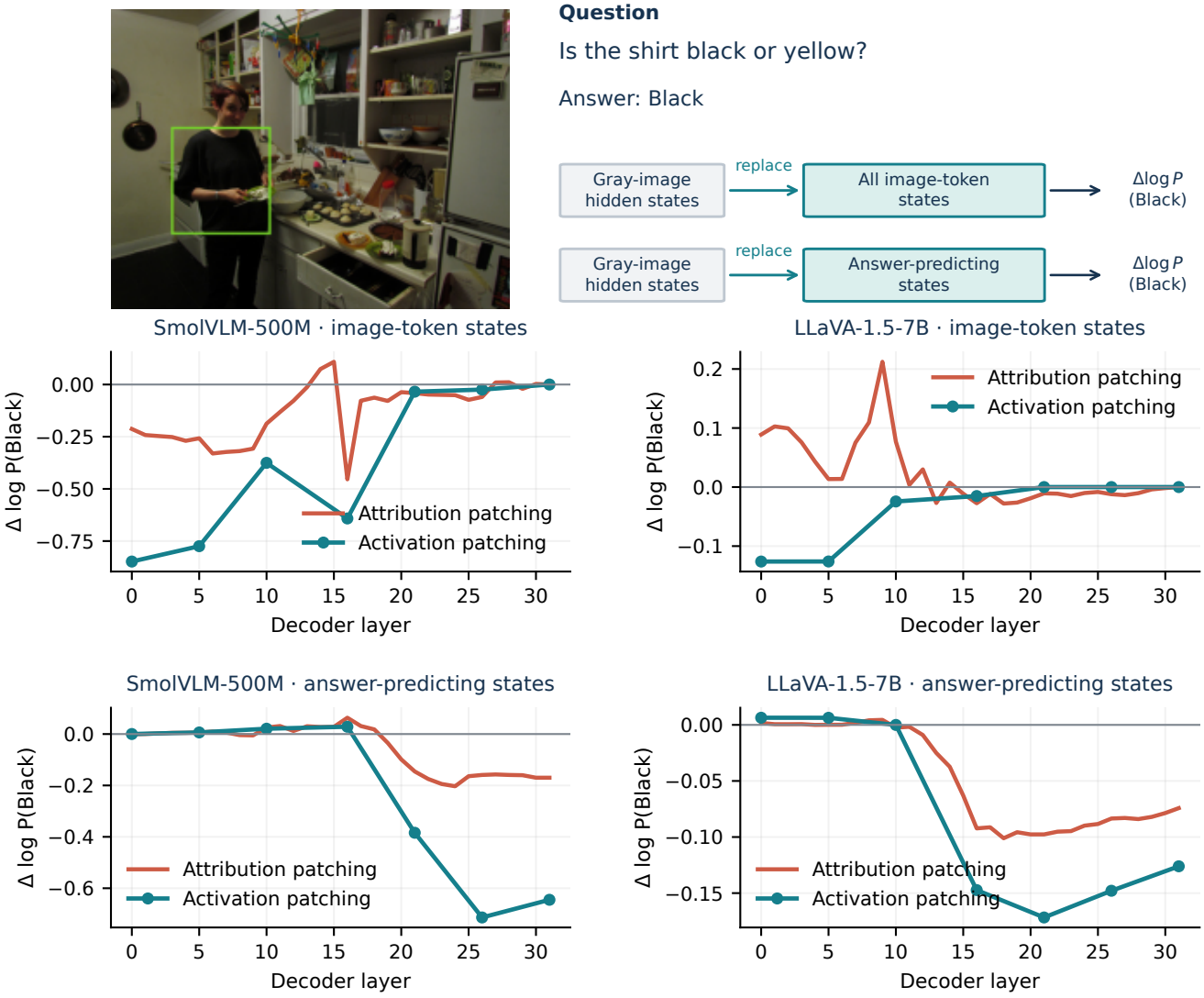



**Figure 20.** Exact and first-order residual replacement for the black-shirt question. LLaVA's measured effects are markedly smaller than on Yellow, whereas SmolVLM remains sensitive to the gray-image reference. Both methods use the same native sites and complete Black score.

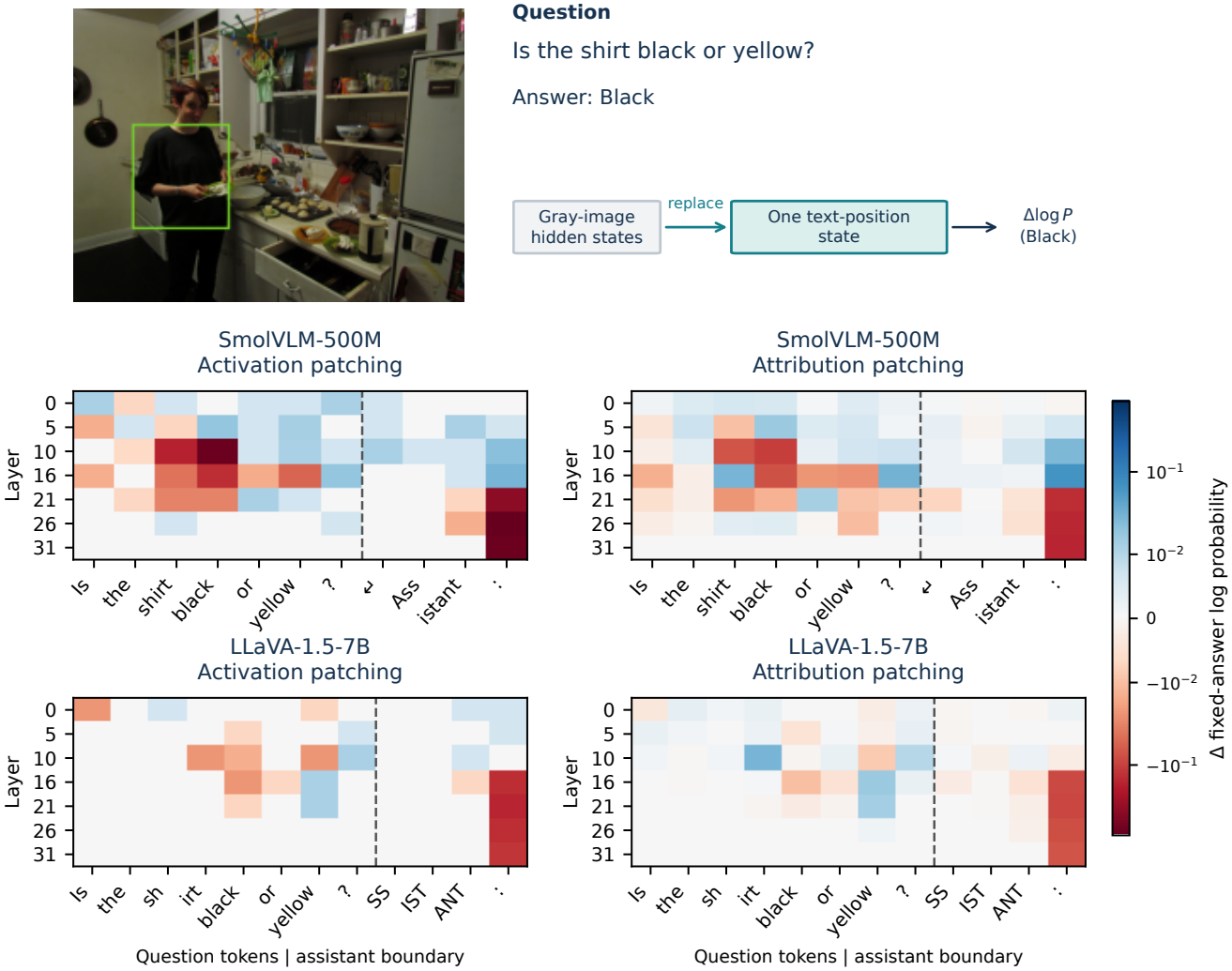



**Figure 21.** Text-position patching for the black-shirt input. Exact replacements identify intermediate question-word and later prediction-position effects; the aligned first-order maps provide a direct approximation check rather than a separate attribution target.